



International Conference on
Plant Health Management

ICPHM 2023: Innovation and Sustainability

Hyderabad, 15-18 November, 2023



Golden Jubilee
(1972- 2022)

SOUVENIR

Organized by

Plant Protection Association of India

National Bureau of Plant Genetic Resources, Regional Station

Rajendranagar, Hyderabad - 500 030, India





International Conference on
Plant Health Management
ICPHM 2023: Innovation and Sustainability
Hyderabad, 15-18 November, 2023

Golden Jubilee
(1972- 2022)

SOUVENIR

Organized by

Plant Protection Association of India

National Bureau of Plant Genetic Resources, Regional Station
Rajendranagar, Hyderabad - 500 030, India





Published by

Dr B Sarath Babu, President,
Plant Protection Association of India

Compiled and Edited by

Dr Gururaj Katti

Dr K Anitha

Dr Prasanna Holajjer

Dr B Bhaskar

Dr B Parameswari

Dr T V K Singh

Printed at

Sai Maruthi Print Solutions, Hyderabad,

Email: saimaruthiprintsolutions@gmail.com

Mobile: 9912277127, 7780688102

SINGIREDDY NIRANJAN REDDY

Minister for Agriculture,
Co-operation & Marketing
Government of Telangana.



Off: 2nd Floor, HACA Bhavan,
Nampally, Hyderabad - 500022.
Phone: 040 - 2345 0813 / 0533
e-mail: niranjan.singireddy@gmail.com

Res: Quarter No.3,
Minister's Quarters, Road No.12,
Banjara Hills, Hyderabad - 34.
Phone: 040 -2956 9119 / 2331 0813

MESSAGE

Agriculture is the backbone of India's economy, with the majority of the population depending on it for their livelihood. In Telangana, the state government's farmer-friendly policies have transformed agriculture, created a positive and enabling ecosystem; and conducive environment to boost the farmers' confidence, making Telangana the seed bowl of India and a model.

The state government has been implementing several policies for assisting the farmers in various ways from providing financial assistance to offering advice and marketing facilities. The expansion of irrigation facilities and renewal of irrigation tanks has led to a significant increase in acreages dedicated to cotton and paddy, resulting in improved farm productivity and enhanced prosperity for farmers. With the implementation of proactive measures to ensure the availability of high-quality agricultural inputs, farmers now have easy access to superior seeds, fertilizers, and agrochemicals.

Plant Protection continues to play a significant role in achieving targets of crops production. The major thrust areas of plant protection are promotion of integrated pest management, ensuring availability of safe and quality pesticides for sustaining crop production from the ravages of pests and diseases, streamlining the quarantine measures for accelerating the introduction of new high yielding crop varieties, besides eliminating the chances of entry of exotic pests and human resource development including empowerment of women in plant protection skills.

Telangana state government has deployed the mobile vans for farmer awareness campaign 'JagoKisanJago' on quality

agrochemicals and proper billing to make quality pesticides available and help the farmers improve their yields. This campaign also aims to integrate pest management techniques into crop management practices, educate farmers on the proper disposal of empty containers, and emphasize accurate timing and methodologies for the application of agrochemicals.

I whole heartedly congratulate the Plant Protection Association of India (PPAI) on completion of 50 years of yeoman service for the cause of Plant Protection both at National and International level and organizing an International Conference on Plant Health Management mainly to pursue for global opportunities in Innovation and Sustainability of research and entrepreneurship relating to the field of plant health management.

Besides, there is need to find ways to build sustainable and inclusive food systems, focussed mainly on small and marginal farmers. We must find ways to strengthen food supply chains. At the same time, adopt agricultural practices for better soil health, crop health and yield. Traditional practices from different parts of the world may be taken into consideration to develop alternatives for regenerative agriculture. There is need to empower our farmers with innovation and digital technology and also make solutions affordable for small and marginal farmers. There is also an urgent need to reduce agricultural and food waste, and instead, invest in creating wealth from waste.

I hope the deliberations in this conference would succeed in bringing out a roadmap for the identification, delineation and exploitation of the core research areas of Plant Health Management for business prospects in the field of Agriculture. Lastly, I wish that this International Conference on Plant Health Management would be a grand success.



(S.NIRANJANA REDDY)



डॉ. हिमांशु पाठक

DR. HIMANSHU PATHAK

सचिव (डायरेक्टर) एवं महानिदेशक (आईसीएआर)

Secretary (DARE) &
Director General (ICAR)

भारत सरकार
कृषि अनुसंधान और शिक्षा विभाग एवं
भारतीय कृषि अनुसंधान परिषद
कृषि एवं किसान कल्याण मंत्रालय, कृषि भवन, नई दिल्ली-110 001

GOVERNMENT OF INDIA
DEPARTMENT OF AGRICULTURAL RESEARCH AND EDUCATION (DARE)
AND
INDIAN COUNCIL OF AGRICULTURAL RESEARCH (ICAR)
MINISTRY OF AGRICULTURE AND FARMERS WELFARE
Krishi Bhavan, New Delhi 110 001

Tel: 23382629 / 23386711 Fax: 91-11-23384773

E-mail: dg.icar@nic.in



Message

I am happy to know that an “International Conference on Plant Health Management ICPHM 2023 - Innovation & Sustainability” is being organized by Plant Protection Association of India at Professor Jayashankar Telangana State Agricultural University (PJTSAU), Hyderabad, Telangana, India during November 15-18, 2023.

The Plant Protection Association of India is sailing through its journey quite admirably in the last few decades, and maintained high standards all through its mandate activities. The conference aims to pursue global opportunities in Innovation and Sustainability of research and entrepreneurship relating to the field of PHM. I am sure the deliberations of this conference will nurture creativity and inspire ecosystem to focus on preparing a roadmap for the identification, delineation and exploitation of the core research areas of PHM for business prospects in the field of Agriculture. I compliment the efforts of the organizing committee and core team for their encouraging and engaging role in keeping the flag of Plant Protection Association of India flying high and for adding value to cause of plant protection by encouraging and promoting plant protection research education and extension activities in the country.

I wish the International conference a grand success.

(Himanshu Pathak)

27th October, 2023
New Delhi

भारतीय कृषि अनुसंधान परिषद
कृषि एवं किसान कल्याण मंत्रालय
भारत सरकार, कृषि भवन
नई दिल्ली 110001, भारत



Indian Council of Agricultural Research
Ministry of Agriculture and Farmers Welfare
Govt. of India, Krishi Bhavan
New Delhi 110001, India

डॉ. तिलक राज शर्मा
उप महानिदेशक (फसल विज्ञान)

Dr. T. R. Sharma, Ph.D
FNA, FNAAS, FNASc, JC Bose National Fellow
Deputy Director General (Crop Science)



MESSAGE

The field of crop protection has evolved from the basic idea of integrated pest control followed by integrated pest management into the present holistic concept of Plant Health Management (PHM). It has opened a new era in safeguarding agri-horticultural crops against the attack of arthropod pests, plant diseases and weeds for the benefit of farming community. In this cause, PPAI has demonstrated its ability to navigate the fruits of scientific research to all the concerned stakeholders through various activities.

I am happy to note that on occasion of their Diamond Jubilee Celebrations, PPAI came forward to organize the "International Conference on Plant Health Management ICPHM 2023 - Innovation & Sustainability" from 15 -18 November 2023. The conference is very aptly designed to provide a platform for exchange of ideas, collaborations and partnerships to address the challenges of increasing human population, global climate change with its impact on agriculture and the relentless scientific conundrum of human made artificial intelligence, that will need urgent attention and redressal.

I take this opportunity to congratulate the PPAI board and wish them a grand success for conduct of the four-day conference that will bring useful recommendations and suggestions, that are needed to be mainstreamed with the ongoing efforts of PPAI that ensure enabling agricultural ecosystem with innovation and sustainability.


(T.R. Sharma)

Place : New Delhi

Dated 26th October' 2023

डॉ. सुनील चन्द्र दुबे
सहायक महानिदेशक
(पादप संरक्षण एवं जैव सुरक्षा)

Dr. S. C. Dubey
Assistant Director General
(Plant Protection and Biosafety)



भारतीय कृषि अनुसंधान परिषद
कृषि एवं किसान कल्याण मंत्रालय, भारत सरकार
क.सं. 215, कृषि भवन, नई दिल्ली, भारत

Indian Council of Agricultural Research
Ministry of Agriculture & Farmers Welfare, Govt. of India
R.No. 215, Krishi Bhawan, New Delhi 110001, India

☎: 91-11-23384414; ☎: 91-11-23384414
✉: adgpp.icar@nic.in; sunil.dubey@icar.gov.in

Date: October 24, 2023

MESSAGE

I am glad to note that an International Conference on Plant Health Management: Innovation and Sustainability (ICPHM 2023) is being organized by Plant Protection Association of India from November 15-18 2023 that aims to bring together entomologists, plant pathologists, nematologists, weed scientists, chemists, agronomists, economists, plant breeders, technology transfer specialists and policy makers to discuss and communicate new science in plant protection for sustainable plant health management. The program of ICPHM 2023 is well designed to address the key issues in plant protection against the backdrop of climate change and mounting pressure on natural resources to meet the growing need for nutritious and safe food, conservation of biodiversity and creating opportunities for economic growth.



The ICPHM 2023 will provide a great opportunity to present, discuss and disseminate the recent advances in plant protection, and establish contact and collaboration between plant protection scientists and students from different parts of the world. The deliberations of ICPHM 2023 will include keynote lectures, plenary speeches, and concurrent sessions in different areas related to plant protection. The venue of the ICPHM 2023 is the historic city of Hyderabad, and the hub of India's modern IT industry, which will provide an appropriate setting and up-to-date technical facilities for the Conference.

I convey my heartiest congratulations to the organizers who are making every effort to make Conference a memorable and professionally rewarding event and wish a grand success.

(S.C. Dubey)

PROFESSOR JAYASHANKAR TELANGANA STATE AGRICULTURAL UNIVERSITY

Administrative Office, Rajendranagar, Hyderabad - 500 030, Telangana State, India

M. RAGHUNANDAN RAO, IAS
Vice Chancellor

Phone : +91 - 40 - 24015122

Fax : +91 - 40 - 24018653

Email : vcpjtsau@gmail.com

Message

According to estimates from the Food and Agriculture Organization of the United Nations (FAO), as much as 40 per cent of the world's agricultural crops are lost to pests, wherein invasive pests alone cost at least \$70 billion annually. Amid concerns of rising temperatures and prolonged rainless periods due to climate change, large-scale attacks of pests and diseases are imminent and preserving plant health is fundamental to achieve the Sustainable Development Goals. Sustaining plant health is an integral part of our approach towards more efficient, inclusive, resilient and sustainable agri-food systems. Innovation in agronomic practices, advent of chemicals for control, and more recently genetic engineering tools, drones and AI technologies have been providing new opportunities for reduction of crop losses.

Novel biologically efficient, environmentally sound, user friendly and economically viable crop protection technologies are required to maintain plant health without compromising on the environmental sustainability. Particularly, to ensure nutritional security and at the same time address climate variability related challenges, novel precision agricultural technologies are required for sustainable & cost-effective agriculture growth. To ensure food security, innovations in agriculture are inevitable, that too at breath taking pace.

The efforts of Plant Protection Association of India (PPAI) in crop protection stewardship, particularly focusing on plant health management are commendable. I congratulate PPAI on completion of Fifty golden years and also for organizing the International Conference on Plant Health management focusing on innovation and discuss and deliberate on building sustainable solutions.

Hope that ICPHM 2023 through these deliberations will work towards building novel plant health management solutions, particularly focusing on combating invasive pests, integrating precision technologies, to suit to small farm holds and wish this conference a grand success.


(M. RAGHUNANDAN RAO)



सी. अचलेन्दर रेड्डी, भा.व.से. (सेवानिवृत्त)
अध्यक्ष

C. Achalender Reddy, IFS (Retd.)
Chairperson



राष्ट्रीय जैव विविधता प्राधिकरण
भारत सरकार

National Biodiversity Authority
Government of India

MESSAGE



I am pleased to know that the Plant Protection Association of India (PPAI) has completed 50 years of its establishment and rendering glorious service to the overall development of Agriculture in India. It is pertinent to state that India has achieved self-sufficiency in food grains production through adoption of modern agricultural practices based on use of chemical fertilizers, pesticides, etc. However, it also resulted in loss of biodiversity, land degradation, pollution of soil, water and air, pesticidal residues in food, decline in food quality, escalating production cost, etc. Further, the total lockdown during COVID – 19 pandemic has also aggravated the loss of biodiversity to a great extent.

Modern agriculture and biodiversity are intimately related and are complementary in nature. Large section of population in rural and tribal areas still depends on local flora and fauna for their food, nutritional requirements and economic security. In view of India's huge population and its economic growth, industrialization and globalization there is an immense pressure on biodiversity and eco-system services. The conservation of biodiversity is very important as it plays a great role in enhancing the livelihoods of millions of people. In order to achieve sustainable development through poverty alleviation, India is all set to achieve Sustainable Development Goals by 2030.

The ongoing International Conference on Plant Health management (ICPHM) – Innovation & Sustainability is therefore highly relevant and timely. The ICPHM would lay a roadmap on finding cost effective, ways and means of bio-utilization and conservation of biological resources. There is a need to give priority to identification, upscaling and out-scaling eco-friendly farming practices to enhance the quality of food without jeopardizing biodiversity including preservation of habitats, conservation of landraces, folk varieties and cultivars, domesticated stocks of breeds of animals and microorganisms. The chronicling of knowledge related to biological diversity is also equally important. Besides, all individuals should follow objectives of Mission Life and activities spearheaded by Hon'ble Prime Minister of India. These include reduction in use of plastics, synthetic pesticides and release of harmful substances into environment and promoting circular economy etc. In order to meet requirements of food, fuel and arrest soil erosion, Agro-forestry with multipurpose tree species should be planted on large scale for achieving self-sufficiency in food, wood and nutritional security. This conference would help in strengthening the eco-system management, augmenting carbon sequestration and mitigation of climate change for sustainable future.

I wholeheartedly appreciate the efforts of all the office bearers of the PPAI for bringing out this valuable compendium which would go a long way in benefitting the multi stakeholders. I also wish the conference a grand success.

Date: 12th October, 2023

(C. Achalender Reddy)
Chairperson

5वां तल, टाइसल बायो पार्क, सीएसआईआर रोड, तरमणि, चेन्नई-600 113, तमिल नाडु, भारत.

5th Floor, TICEL Bio Park, CSIR Road, Taramani, Chennai - 600 113, Tamil Nadu, India.

☎+91 44 2254 1085 ☎+91 44 2254 1073 ✉chairman@nba.nic.in : achal.reddy@gmail.com 🌐www.nbaindia.org

त्रिलोचन महापात्र

अध्यक्ष

पौधा किस्म और कृषक अधिकार संरक्षण प्राधिकरण

(संसद के अधिनियम द्वारा निर्मित सांविधिक निकाय)

कृषि एवं किसान कल्याण मंत्रालय

भारत सरकार



सत्यमेव जयते

Trilochan Mohapatra

Chairperson

Protection of Plant Varieties and Farmers' Rights Authority

(A Statutory body created by an Act of Parliament)

Ministry of Agriculture and Farmers Welfare

Government of India

**MESSAGE**

I am pleased to know that the Plant Protection Association of India (PPAI) is organizing an International Conference on Plant Health Management ICPHM 2023 – Innovation & Sustainability at Professor Jayashankar Telangana State Agricultural University (PJTSAU), Hyderabad, Telangana, India from 15-18 November, 2023. I have been informed that this is the 50th anniversary of PPAI and therefore, it is most befitting that the ICPHM is being organized to celebrate the Golden Jubilee of the Association.

The international conference with the tagline “Innovation & Sustainability” in the context of plant health management appears appropriate. Enhancing agricultural production to meet the ever-increasing demands of the growing population has been possible partly because of large-scale use of agrochemicals to manage plant health. This has led to many negative consequences. It is therefore very pertinent to intensify efforts on innovations that would promote sustainable agriculture by minimizing the harmful effects of agrochemicals, and by developing and deploying alternative technologies /approaches in order to ensure long-term sustainability of agri-food system. Engaging youth to build new entrepreneurship and productive partnerships in the wake of global climate change is of paramount importance. While the conference commemorates the completion of 50 golden years of service by PPAI, it should provide a platform to effectively engage farmers, researchers, state and central government agencies as well as other public and private stakeholders in a serious debate and elaborate discussion on issues related to plant health management.

Our vision should be to create even greater value for all players in agri-value chains through timely and appropriate science-based solutions after through experimentation, deliberation, exchange and spread of new knowledge in the areas of plant protection. I do hope that the conference will deliver a set of recommendations regarding R&D, building productive partnerships and networks, addressing policy imperatives and reaching the unreached for sustainable management of plant health.

I wish the conference a grand success.


(Trilochan Mohapatra)Date: 30th October, 2023

एन ए एस सी काम्लैक्स, डी पी एस मार्ग, नई दिल्ली-110012

NASC Complex, DPS Marg, New Delhi-110012

Tel: 011-25848127, 25841696; Mob.: +91-7840000542; Fax: 011-25840478

Website: www.plantauthority.gov.inE-mail: chairperson-ppvfra@nic.in



National Institute of Plant Health Management

Department of Agriculture & Farmers Welfare
Ministry of Agriculture & Farmers Welfare
Government of India



राव स्वा प्रस
NIPHM

Dr. SAGAR HANUMAN SINGH, IPoS
Director General
Tele-Fax: +91-40-24015346
E-mail: dgniphm@nic.in

Rajendranagar,
Hyderabad - 500030
<http://niphm.gov.in>



Hyderabad
13.11.2023

Message

With immense pleasure and pride I heartily congratulate the Plant Protection association of India for organizing the International Conference on Plant Health Management-ICPHM-2-23: Innovations & Sustainability on 15-18th Nov 2023, at PJTSAU campus, Hyderabad.

The Indian Agriculture is experiencing a wide range of challenges such as climate change, excessive dependence on chemical agriculture threat is on invasive pest and post-harvest losses in both Agriculture as well as Horticulture because of improper plant protection measures on the other hand there is host of opportunities for the Indian Farmers to explore the use of Indian Technical Knowledge (ITK) in ameliorating both in the soil as well as the plant health on one hand and exporting the surplus Agri/Horticulture produce to various countries of the world. Thus provides additional income to the farmers, in addition to the providing food and nutritional security to our own citizenries our country has released agriculture/horticulture production in the year 2022-2023 and the world looks up to India to provide food security to Agricultural deficient countries. In this background the content International Conference on Plant Health Management bears the torch to highlight the areas which need to be underlined for the benefit of policy makers, administrators, researchers, academia and the farming community to urgently tackle the key challenges which threaten food and nutritional security of this country.

I hope this conference will through adequate light of issues such as new recent trance in the Plant Health Management threatening the issues, threatening the Plant Biosecurity, putting adequate infrastructure for Plant Protection different states of the country and key issues of capacity building in Plant Health Management. I wish the organisers and all the stake holders of the directly or indirectly involved in ICPHM 2023 all success.



[S H Singh]



Dr R. Sarada Jayalakshmi Devi
Hon'ble Vice Chancellor

ACHARYA N.G. RANGA AGRICULTURAL UNIVERSITY

Administrative Office

Lam, Guntur – 522 034, A.P, India

Website: www.angrau.ac.in



Message

Global report on food crisis (GRFC) indicated that, in the year 2022 around 258 million population in 58 countries and territories are facing acute food insecurity. Climate change impacts are contributors to decreased crop yields, accentuation of invasive pests, and shifting agro-ecosystem boundaries. As the global population continues to grow, food demand of 70-100% increase is expected by 2050, putting pressure on agricultural land. Achieving food security, one of the UN Sustainable Developmental Goals (SDGs) is a significant challenge, and addressing factors threatening the sustainability of agricultural production systems is imperative to mitigate the threat posed by global climate change. Another major challenge has been implementing efficient, environmentally safe crop protection measures to save 40% of crop losses amounting to Rs. 50,000 crores annually. However, use of pesticides has remained as one of the major strategies and the global crop protection chemicals market has been estimated to be worth \$63.7 billion in 2020 and is poised to reach \$74.1 billion by 2025, despite availability of several eco-friendly and integrated approaches to manage several insect pests and diseases.

Climate smart technologies, biotechnological innovations and emerging technologies like IOTs, drones, AI and Digital agriculture and biologically sustainable options need to be integrated under precision agriculture capsule to ensure farmers productivity and profitability. Unravelling or harnessing science through exploring new avenues and plant health management strategies shall also focus towards not only plant health management but also creation of rural employment opportunities.

Plant Protection Association of India (PPAI), Hyderabad has been playing a pivotal role in promoting such novel strategies since fifty golden years and I congratulate PPAI on completion of 50 fruitful years.

Hope that this, International Conference on Plant Health Management (ICPHM-2023) shall bring a paradigm shift in evolving sustainable crop protection solutions and wish ICPHM 2023 a grand success.


(R. SARADA JAYALAKSHMI DEVI)



Dr. (Smt.) B. Neeraja Prabhakar
 M.Sc. (Ag.) Horti., Ph.D.
 Vice – Chancellor &
 Chairman, RAC, ICAR-IIPR.



SRI KONDA LAXMAN TELANGANA STATE
 HORTICULTURAL UNIVERSITY

Administrative Office,
 Mulugu (Village & Mandal),
 Siddipet District - 502279
 Email: vcskltshu@gmail.com
vc@skltshu.ac.in
 Mobile: +91 8333981351

SOUVENIR MESSAGE

Indian Horticulture sector has been the mainstay of Indian Agriculture with a contribution of about 30 per cent to the agricultural GDP from about 14 per cent area and 40 per cent of total export earnings in agriculture as a whole. Horticulture production increased 13-folds from 25 million metric tons during 1950-51 to 342.30 million tonnes during 2021-22 surpassing food grain production. According to FAO (2022), amongst fruits, the country ranks first in the production of Banana, Mango, Guava & Papaya while amongst vegetables, first in producing Okra, Onion & Shallots, Chillies & Beans and second in the Potatoes, Tomato, Cauliflowers, Broccoli, Brinjal, Cabbages and Green peas.

According to FAO, in the foreign trade, export growth of fresh fruits and vegetables is valued at 14 per cent and of processed fruits and vegetables at 16.27 per cent. On an average, pests account for 20-40% of yield losses worldwide, costing the global economy a combined \$290 billion.

Due to the impact of climate change, plant pests that ravage economically important crops are becoming more destructive and posing an increasing threat to food security and the environment. In spite of having stringent quarantine regulation, the invasive alien species often get established. In Telangana, during the last five years Tomato leaf miner *Tuta absoluta* (Meyrick) (Lepidoptera: Gelechiidae) was observed on tomato in 2014, Rugose spiraling whitefly *Aleurodicus rugioperculatus* Martin (Hemiptera: Aleyrodidae) was observed on oil palm and Coconut in 2017, Bondar's nesting whitefly *Paraleyrodobondari* Peracchi (Hemiptera: Aleyrodidae) was observed on oil palm and Coconut in 2019. The recent outbreak of thrips, *Thrips parvispinus* (Kudo) in chilli fields of Telangana and Andhra Pradesh during 2021 has alerted the farmers and other stakeholders at large.

Although application of pesticides has helped mitigation of plant diseases and insect pests, their indiscriminate use causes environment pollution, chemical residues above threshold levels and emergence of new races/biotypes. Therefore, breeding for the development of biotic resistant cultivars in horticultural crops, through identifying novel genes conferring pest and disease resistance coupled with comprehensive studies on host-pathogen interactions greatly improved the crop protection efficiency and also resulted in the development of several high-yielding genetically resistant cultivars.

Innovative genetic engineering approaches, automation of plant health monitoring, use of drones either for pest detection and diagnosis or pesticide application would bring in sea change in evolving efficient plant protection solutions. In this context, Plant Protection Association of India (PPAI) is striving hard, since 50 years to create awareness among researchers on such novel plant health management strategies.

Their unstinted efforts are commendable and I wish that the International Conference on Plant Health Management (ICPHM, 2023) would focus on finding new solutions not only to combat existing crop pests but also keep an eye on impending threats from invasive pests. I congratulate team PPAI and wish ICPHM 2023 a grand success.


 B. NEERAJA PRABHAKAR
 VICE CHANCELLOR



Dr.Y.S.R. Horticultural University

www.drysrhu.ap.gov.in

Dr.T.JANAKIRAM, ARS

Ph.D. (IARI), FISOH, FISVS, FHSI & FISHRD

Vice-Chancellor



Message

The FAO projections show that feeding a world population of 9.1 billion people in 2050 would require raising overall food production by about 70 percent between 2005 and 2050. That means, production in the developing countries would need to almost double. To meet the demands of rapidly increasing population from a limited land resource the use of intensive agri-horti farming systems has already lead to the accentuation of pests and diseases. Coupled with the consequences of climate change, currently we are witnessing either increased host plant susceptibility or invasive pests posing threat to crop production. To reduce the use of pesticides, the time has come to look for eco-friendly and novel solutions for pest management. Apart from several existing crop protection options such as biofumigation, plant defense activators, pathogenesis-related proteins, biointensive IPM, fusion protein-based biopesticides and use of semiochemicals, emerging cutting-edge technologies like Crisper CAS, nanotechnology, drones, IOT, AI&ML based devices, robotics coupled with automation and digitization need to be explored to arrive at profitable and sustainable plant health management solutions.

Therefore, forging ahead to promote holistic plant health management concept, would open up several new avenues for researchers across the globe through intense debates and deliberations. In this endeavour, the Plant Protection Association of India (PPAI) has demonstrated its ability and stood in the forefront to encourage researchers of crop protection not only to publish through IJPP but also engaging them in discussions through organizing several conferences and symposia, to further the cause of plant protection since fifty golden years.

Organizing the "International Conference on Plant Health Management (ICPHM 2023) – with focus on Innovation & Sustainability" in the current scenario of looming threats of global food insecurity and dwindling farm profitability is highly commendable. I wish that this very significant event would be successful and fruitful.



Vice Chancellor

Administrative Office, Venkataramannagudem - 534101, West Godavari District, Andhra Pradesh, India.

Ph : 08818-284312, Mobile : 7382633666, Email : vc@drysrhu.edu.in



FEDERATION OF SEED INDUSTRY OF INDIA

10A, 10th Floor, Vandhana Building, 11 Tolstoy Marg, New Delhi -110001.
Website: www.fsii.in | CIN:91990DL2016NPL306449



MESSAGE

Greetings from the Federation of Seed Industry of India (FSII), an association of research based seed companies in India.

The Plant Protection Association of India is sailing through its journey quite admirably in the last few decades, and maintained high standards all through its mandate activities. I am delighted to note that an “International Conference on Plant Health Management ICPHM 2023 - Innovation & Sustainability” is being organized by Plant Protection Association at Professor Jayashankar Telangana State Agricultural University (PJTSAU), Hyderabad, Telangana, India from 15 -18 November 2023.

I understand the conference aims to pursue global opportunities in Innovation and Sustainability of research and entrepreneurship relating to the field of PHM. As agriculture is the backbone for the economy of many countries and majority of the human population dependent on it for food, feed, and fibre, the chosen topic of this International Conference, Plant Health Management, is a significant necessity in the current scenario. I am sure the deliberations of this conference will nurture creativity and inspire the ecosystem to focus on preparing a roadmap for the identification, delineation and exploitation of the core research areas of PHM for business prospects in the field of Agriculture.

On behalf of FSII, I compliment the efforts of the organizing committee and core team for their encouraging and engaging role in keeping the flag of Plant Protection Association of India flying high and for adding value to cause of plant protection by encouraging and promoting plant protection research education and extension activities in the country.

I wish the International conference a grand success.

Date: 20th Oct 2023

Ram Kaundinya
Director General



Rahul Dhanuka
Joint Managing director



Message

On the behalf of Dhanuka Group, I express my heartfelt congratulations to the Plant Protection Association of India (PPAI) on the completion of 50 years of service and convey my best wishes for continued excellence. I also wish (PPAI) a grand success for their “International Conference on Plant Health Management – Innovation & Sustainability” (ICPHM 2023) being organized at Hyderabad, India during Nov 15 to 18, 2023.

I am sure that this forum will allow farmers, academics, state and federal government agencies, and all stakeholders to engage in serious introspection and discussion about plant health management challenges.

Understanding the interaction of biotic and abiotic factors that prevent economically significant plants from reaching their full genetic potential plant health management formally impacts cereal crops, horticulture crops, even forest flora & other economically significant plants is considered under the practice of plant health management (PHM).

PHM centers on the idea of Integrated Pest Management (IPM) of these aforesaid stresses to provide benefits to the environment, economy, and society as well as for all the stakeholders involved.

Through this event ICPHM-2023 is offering a platform for exchanging ideas on innovation and sustainability in the domain of plant health management research and entrepreneurship, while also addressing global climate change and challenges in this field.

Organizing this conference will undoubtedly contribute in the discovery, demarcation, and utilization of primary PHM research topics for innovation and sustainability prospects in the agriculture sector.

My best wishes for the grand success of ICPHM 2023.

Gurugram,
06 November 2023

(Rahul Dhanuka)
Joint Managing Director
Dhanuka Agritech Ltd.,
Gurugram-122002, Haryana, India

Organizing Committee of ICPHM 2023

Chief Patron
Shri Narendra Singh Tomar, <i>Honourable Union Minister of Agriculture and Farmers Welfare, Government of India, President of the ICAR Society</i>
Patron
Shri Singireddy Niranjan Reddy, <i>Honourable Minister of Agriculture, Co-Operation and Marketing, Government of Telangana</i>
Dr Himanshu Pathak, <i>Secretary, DARE, Govt. of India & Director General, ICAR, New Delhi</i>
Dr R S Paroda, <i>Founder Chairman, Trust for Advancement of Agricultural Sciences (TAAS) & Former Secretary DARE & Director General, ICAR, New Delhi</i>
Dr Trilochan Mohapatra, <i>Chairperson, PPVFRA, New Delhi</i>
Shri C Achalender Reddy, <i>Chairperson, National Biodiversity Authority, Chennai</i>
Dr V Praveen Rao, <i>Ex Vice-Chancellor, PJTSAU, Hyderabad</i>
Conference Chair
Dr B Sarath Babu, <i>President, Plant Protection Association of India, Former Principal Scientist & Head ICAR-NBPGR RS, Hyderabad</i>
Conference Co-Chair
Dr Rajan Sharma, <i>Cluster Leader – Crop Protection and Seed Health, ICRISAT, Patancheru, Hyderabad</i>
Dr R Jagadeeshwar, <i>Former Director of Research, PJTSAU, Hyderabad</i>
Dr V Celia Chalam, <i>Head & Principal Scientist, Division of Plant Quarantine, ICAR-NBPGR, New Delhi</i>
Prof T V K Singh, <i>Dean of Agriculture PJTSAU (Retd.), Ex. Emeritus Scientist-ICAR</i>
Dr Gururaj Katti, <i>Former Principal Scientist (Entomology) & Head (Crop Protection), ICAR – IIRR, Hyderabad</i>
Dr S J Rahman, <i>Senior Professor & University Head of Entomology, Department of Entomology, College of Agriculture, PJTSAU, Hyderabad</i>
Dr M Srinivasa Prasad, <i>Principal Scientist (Plant Pathology) & Head (Crop Protection) ICAR-IIRR, Hyderabad</i>
Conference Secretary
Dr B Parameswari, <i>Senior Scientist, ICAR-NBPGR RS, Hyderabad</i>

Advisory Committee

International Institutions
Dr Shoki Al-Dobai , Team Leader “Locusts and other Transboundary Plant Pests and Diseases, Food and Agriculture Organization (FAO), Rome, Italy
Dr S B Sharma , President, The World Bio Protection Forum, Adjunct Professor, Harry Butler Institute, Murdoch University, WA, Australia
Dr R K Khetarpal , Executive Secretary, APAARI, Bangkok
Dr Hanu R Pappu , Professor, Washington State University, Pullman, Washington, USA
Dr Ravindra Chandra Joshi , Senior Consultant, Philippine Rice Research Institute, Philippines
Dr Hari Kishan Sudini , Principal Scientist (Groundnut Pathology), ICRISAT, Patancheru, Hyderabad
Dr Niranjan M Kumar , CEO, ABS Inc. NJ, USA
Dr Sudeep Bag , Asst. Professor, College of Agricultural & Environmental Sciences, University of Georgia, USA
Dr Keshav Kranthi , Chief Scientist, ICAC, Washington, USA
Dr Mark Hoddle , Professor in Biological Control, University of California, Riverside, USA
Dr Hamadttu Abdel Farag El-Shafie , Associate Professor of Entomology, Date Palm Research Center, King Faisal University, KSA
Dr Karim Maredia , Professor & Associate Dean, College of Agriculture and Natural Resources, Michigan State University, USA
Dr J R Faleiro , Senior Consultant & Former Plant Protection Officer, FAO of the UN
Dr Kuldeep Singh , Cluster Leader - Genebank Head, ICRISAT, Patancheru, Hyderabad
Dr J C Rana , Country Director, Alliance of Bioversity International and CIAT - India, NASC Complex, New Delhi 110012
Public Sector Institutions
Shri C Achalender Reddy , Chairman, National Biodiversity Authority, Chennai
Dr Tilak Raj Sharma , DDG (Crop Science), ICAR, New Delhi
Dr Sagar Hanuman Singh , Director General, NIPHM, Hyderabad
Dr S C Dubey , ADG (Plant Protection & Biosafety), ICAR, New Delhi
Dr P K Ghosh , Director and Vice-Chancellor, ICAR – NIBSM, Raipur
Dr A Vishnuvardhan Reddy , Former Vice-Chancellor, ANGRAU, Guntur
Dr B Neeraja Prabhakar , Vice-Chancellor, SKLTSHU, Mulugu, Hyderabad
Dr T Janakiram , Vice-Chancellor, Dr YSR Horticultural University, Venkataramannagudem
Dr Vinod Kumar Singh , Director, ICAR - CRIDA, Hyderabad
Dr Subhash Chander , Director, ICAR– NCIPM, New Delhi
Dr S N Sushil , Director, ICAR– NBAIR, Bengaluru

Dr R M Sundaram , <i>Director, ICAR– IIRR, Hyderabad</i>
Dr R K Mathur , <i>Director, ICAR– IIOR, Hyderabad</i>
Dr K S Varaprasad , <i>Former Director, ICAR - IIOR, Hyderabad</i>
Dr (Mrs) C Tara Satyavathi , <i>Director, ICAR - IIMR, Hyderabad</i>
Dr Alok K Srivastava , <i>Director, ICAR - NBAIM, Mau</i>
Dr Y G Prasad , <i>Director, ICAR-CICR, Nagpur</i>
Dr V V Ramamurthy , <i>President, Entomological Society of India, New Delhi</i>
Dr R Viswanadhan , <i>Director ICAR-IISR, Lucknow</i>
Dr R Selvarajan , <i>Director, ICAR-NRC Banana, Tiruchirapalli</i>
Dr G P Singh , <i>Director, ICAR-NBPGR, New Delhi</i>
Dr Mukesh K Dhillon , <i>Division of Entomology, ICAR-IARI, New Delhi</i>
Dr J S Bentur , <i>Subject matter specialist, Agri Biotech Foundation, Hyderabad</i>
Dr Dilip Ghosh , <i>ICAR-CCRI, Nagpur</i>
Dr B M K Reddy , <i>Chairman, A P Biodiversity Board</i>
Private Sector Institutions
Mr M Prabhakar Rao , <i>Chairman & Managing Director of Nuziveedu Seeds Pvt. Ltd</i>
Dr V Satyadev , <i>Senior General Manager, R&D, Advanta India Ltd</i>
Dr T Ramesh Babu , <i>Director, Agriculture and Horticultural Science Department, Vignan University, Vadlamudi</i>
Dr Sudheer Daniel , <i>Research Logistic Lead, Corteva AgriScience</i>
Dr Sairam Reddy Palicherla , <i>Co-Founder, Urban Kisaan</i>
Dr Ranjan Ponnappa , <i>Breeding Coordinator for Okra & Gourds, H M Clause India Pvt Ltd</i>
Dr Ramesh Kaliyaperumal , <i>Technical Lead, Coromandel International Ltd.</i>
Mr Ram Kaundinya , <i>Director General, FSII, New Delhi</i>
Dr Kranthi Kumar , <i>Biosafety Lead, Bayer Crop Science, Hyderabad</i>
Dr A Raja Reddy , <i>Dean of Agriculture, School of Agricultural Sciences, Mallareddy University, Hyderabad</i>
Dr B V S Satish , <i>Sr AVP & Head – Formulation Sales & Marketing, Coromandal International Ltd.</i>
Dr Ramesh Kaliaperumal , <i>Regional PMO for India & Asia, Regulatory, Research and Development, UPL Limited, Bandra (East) Mumbai</i>
Shri Venkateswara Rao Yaganti , <i>Director & CEO, CORNTECH SEEDS PVT LTD, Plot No. 18, H. No. 8 - 63/1, NCL Enclave South, Medchal Road, NH - 44, Secunderabad.</i>
Dr K R K Reddy , <i>Managing Director, Sri BioAesthetics Pvt Ltd, Saanvi Classic, First Floor, Plot no. 50, Jayabheri Enclave, Gachibowli, Hyderabad</i>

Core Committee

Dr B Sarath Babu, <i>Chair-ICPHM 2023, President-PPAI, Hyderabad</i>
Dr B Parameswari, <i>ICAR-NBPGR RS, Hyderabad</i>
Dr Sathyanarayana Jella, <i>Dean of Student Affairs, PJTSAU, Hyderabad</i>
Dr T Ramesh Babu, <i>Director, Agriculture and Horticultural Science Department Vignana University, Vadlamudi</i>
Dr A Raja Reddy, <i>Dean of Agriculture, School of Agricultural Sciences, Mallareddy University, Hyderabad</i>
Dr J S Bentur, <i>Subject Matter Specialist, Agri Biotech Foundation, Hyderabad</i>
Dr Alice R P Sujeetha, <i>Director (PBD), NIPHM, Hyderabad</i>
Dr Anitha Kodaru, <i>Principal Scientist & Former Head, ICAR-NBPGR RS, Hyderabad</i>
Dr Bhaskar Bajaru, <i>Scientist, ICAR-NBPGR RS, Hyderabad</i>
Dr V Prakasam, <i>Senior Scientist, ICAR-IIRR, Hyderabad</i>
Dr B S Gotyal, <i>Senior Scientist, ICAR-NBAIR, Bangalore</i>
Dr C Narendra Reddy, <i>Associate Dean, College of Agriculture, Hyderabad</i>
Dr Prasanna Holajjer, <i>Senior Scientist, ICAR-NBPGR RS, Hyderabad</i>
Dr D Sagar, <i>Scientist, Division of Entomology, ICAR-IARI, New Delhi</i>
Dr L Saravanan, <i>Senior Scientist, ICAR-NBPGR RS, Hyderabad</i>
Dr K Rameash, <i>Principal Scientist, ICAR-CICR RS, Coimbatore</i>
Dr J Stanley, <i>Senior Scientist, ICAR-IIMR, Hyderabad</i>
Dr Srinivas Parimi, <i>Bayer Science Fellow, Regional Resistance Management Lead, APAC, Bayer Crop Science India.</i>
Mr A Surender, Ms K Madhavi and Ms Y Prasanthi, <i>PPAI Associates</i>

Local Organizing Committees

Committee	Chairperson	Convenor	Members
Registration Committee	Dr T Uma Maheshwari	Dr D Krishnaveni,	Drs P Lakshmi Soujanya, V Sailaja, B Bhaskar, K Manasa
Accommodation Committee	Dr L Saravanan	Dr S Upendhar	Drs J Stanley, T Boopathy, A Maria Doss, V Prakasam
Food and Catering	Dr P Rajanikanth	Dr M Venkataiah	Drs P Satish, Y Sreedhar, P Duraimurugan
Dias and Decoration Committee	Dr M Venkatshwar Reddy	Dr Ravindra Naik	Dr A Nirmala
Transport Committee	Dr R Sudhakar	Dr K Sakthivel,	Dr K Basawaraj
Memento, awards and certificates printing Committee	Dr A P Padma Kumari	Dr V Sunitha,	Drs Angala Padmasri, Satish Namdeo Chavan, Mallaiah
Technical and Posters Committee	Dr C Narendra Reddy	Drs K Vijaya Laxmi, A P Padma Kumari, Chitra Shanker, N Rama Gopala Varma	Drs V Ravinder Reddy, D Anitha Kumari, R Sunitha, K Kavitha, K Vanisree, Ramya Vittal, P Bindu Priya, T Kiran Babu, D Ladhakshmi, S Srinivasa Reddy, S Om Prakash, K Rama Krishna Babu, I Arunasree
Cultural events Committee	Dr Vidya Sagar	Dr S Triveni	-
Finance Committee	Dr Jella Satyanarayana	Dr S J Rahman,	Drs Dr B Sarath Babu R Jagadeeshwar, Gururaj Katti, Rajan Sharma, R Mahendra Kumar, Y Sreedhar, B Bhaskar
Souvenir / Abstracts and Publication Committee	Dr K Anitha	Dr Gururaj Katti	Drs Parameswari, V Prakasam, P Humayun, G Sridevi, Chitra Shanker, P G Padmaja, T V Prasad, Prasana Holajjer, N Somasekhar, Mool Chand, M Madhavi
Press and Publicity Committee	Dr J Suresh	Dr V Ravindra Naik	Dr K Aruna

Plant Protection Association of India (PPAI) - Fifty Years

Brief History

The Plant Protection Association of India (PPAI) is a scientific society, established in December 1972 at the Central Plant Protection and Training Institute (CPPTI). It is a non-profit organization, registered under the Andhra Pradesh (Telangana Area) Public Societies Registration Act, 1350 Fasli (Act I of 1350 F.) on 4th December, 1972. Its membership is open to all those scientists interested in promotion of Plant Protection and is dedicated for that cause. As per the 1972 memorandum, it was decided that the permanent headquarters of PPAI would be Hyderabad. The Association started functioning from the location of Central Plant Protection Training Institute under the first Presidentship of Dr K K Nirula, the then Director of CPPTI. The office of PPAI was set up in the premises of the institute. During the year 2000, the office set up in its entirety, along with its few office belongings and the balance meager funds were shifted to ICAR - National Bureau of Plant Genetic Resources Regional Station, Hyderabad, with its functional Executive Council drawn mainly from ICAR-NBPGR, Hyderabad. Since then PPAI has been continuing to operate from ICAR-NBPGR, Hyderabad. At this juncture, PPAI has always remained grateful for the services and support rendered by the scientific and other supporting staff for its functional resurrection and new facelift to the Indian Journal of Plant Protection (IJPP) under the stewardship of Dr Y L Nene. The backlog issues of IJPP, since 1997, have been published by the new Editorial Team and



New PPAI building

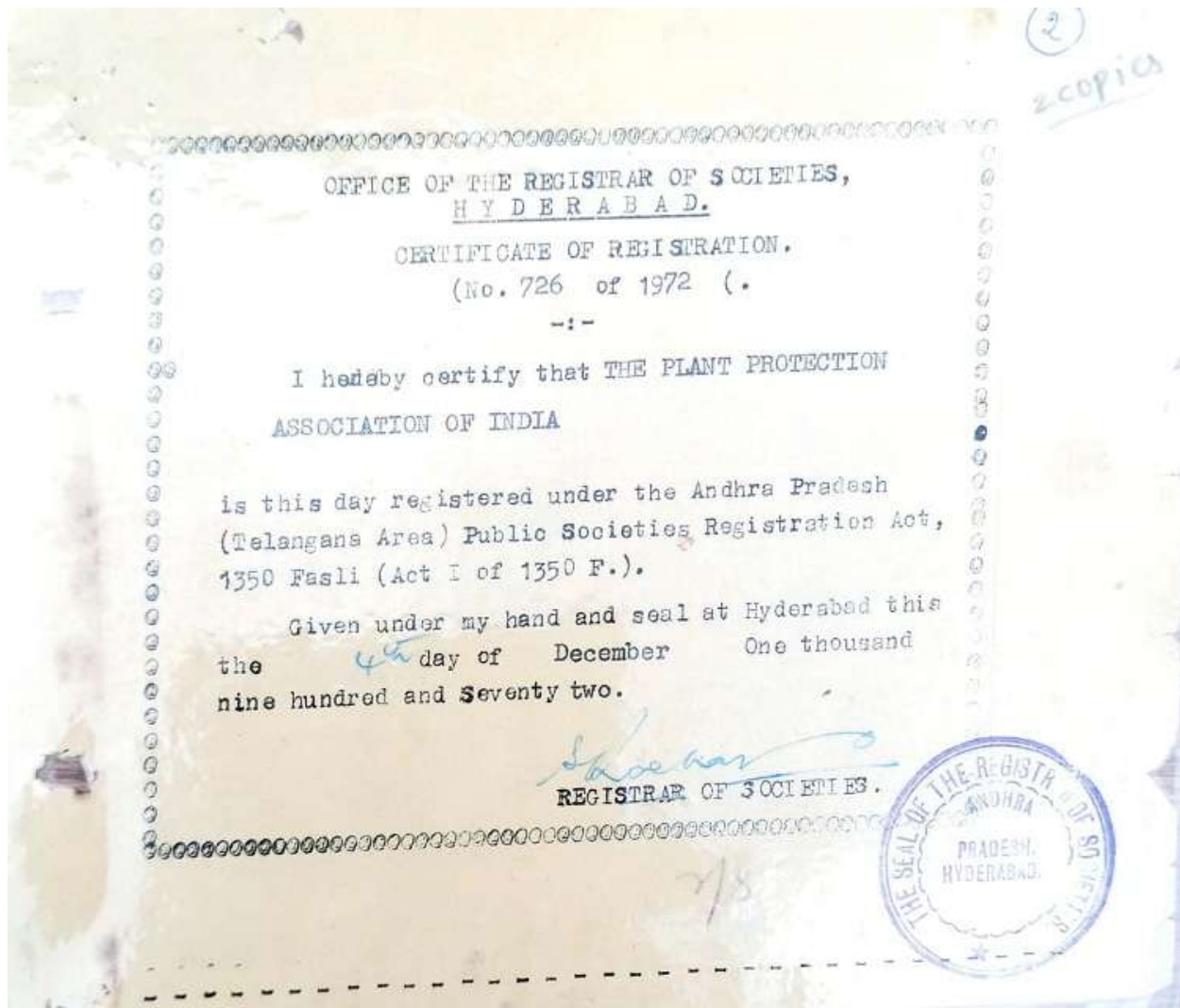
the Journal itself was converted from half yearly to quarterly with new get up leading to meteoric rise of its popularity and its NAAS rating from very low to substantially higher level. In 2007, PPAI was allocated a piece of land in the then Acharya NG Ranga Agricultural University campus (presently PJTSAU Campus, Hyderabad, near Borlaug Guesthouse) and a new one floor building was constructed with the support of few donors. The building was also face lifted to accommodate some of its office material and functional space for the present Executive, led by Dr B Sarath Babu.



Old CPPTI building



ICAR-NBPGR RS Hyderabad building



Certificate of Registration

PLANT PROTECTION ASSOCIATION OF INDIA

MEMORANDUM

- 1 NAME: THE PLANT PROTECTION ASSOCIATION OF INDIA
- 2 HEADQUARTERS: The permanent headquarters of the Association shall be at HYDERABAD
- 3 AIMS & OBJECTIVE: The aims and objectives of the Association shall be:
 - I (a) To maintain a link between plant protection workers of India
 - (b) Officer-trainees & the Central Plant Protection Training Institute for exchange of views, information and establishing close cooperation for the betterment of plant protection in India
 - II (a) To advance the cause of plant protection by encouraging and promoting plant protection research and education in the country
 - (b) disseminating knowledge of plant protection and its application and
 - (c) facilitating closer association among members and between them and other scientific workers in India & abroad

4 DECLARATION:

It is hereby declared that we are desirous of getting the Association registered under the Public Societies Registration Act, 1350 P.

M. A. M. (B. N. Sharma)
Senior Entomologist
CPPTI, Rajawade Nagar, Hyderabad

V. Lakshminarayana (V. Lakshminarayana)
Senior Chemist
CPPTI, Rajawade Nagar, Hyderabad

K. Lakshminarayana (K. Lakshminarayana)
Senior Chemist
CPPTI, Rajawade Nagar, Hyderabad

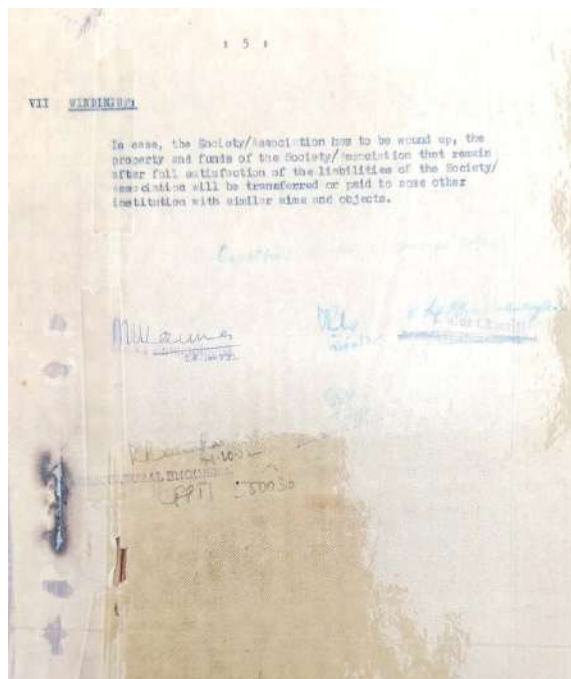
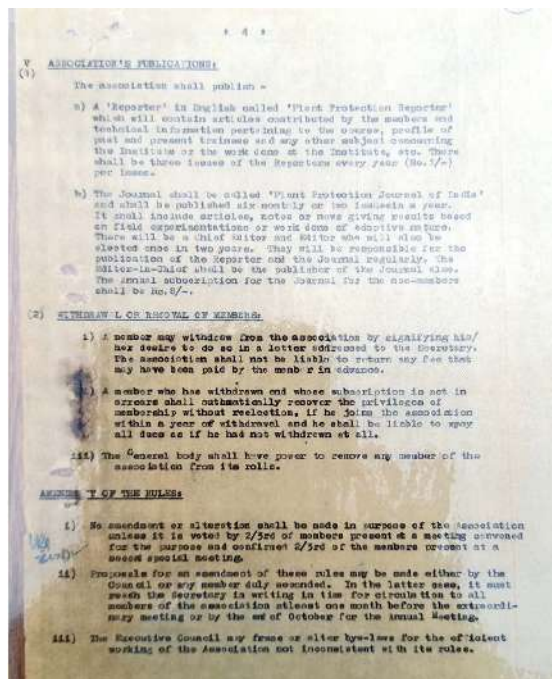
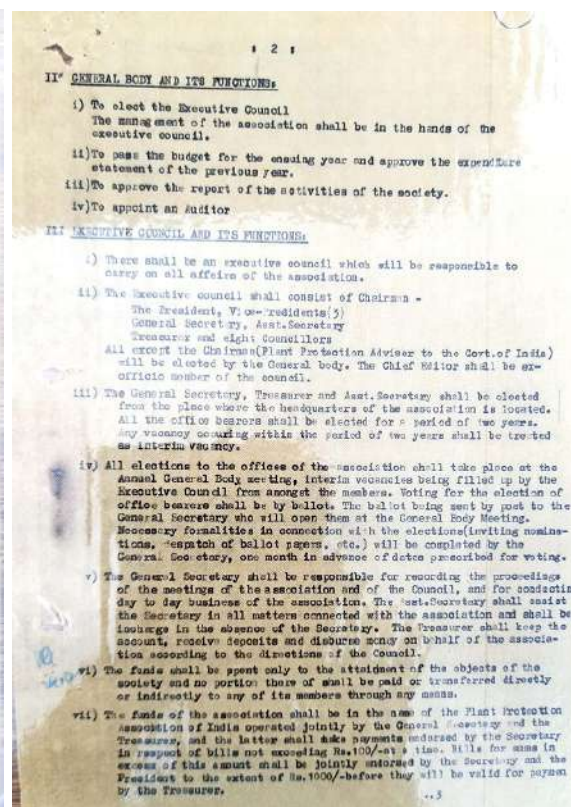
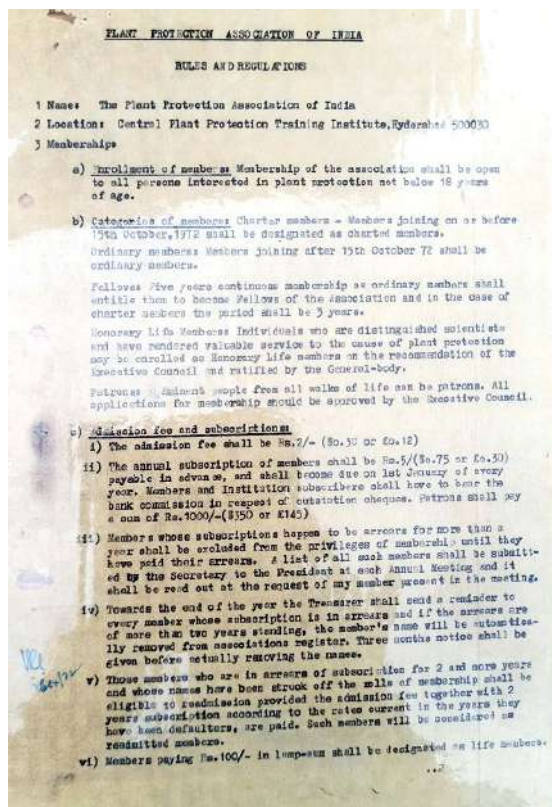
V. S. S. (V. S. S.)
Senior Plant Pathologist
CPPTI, Rajawade Nagar, Hyderabad

S. S. S. (S. S. S.)
A. N. WITNESSES:

P. S. S. (P. S. S.)
A. N. WITNESSES:

S. S. S. (S. S. S.)
A. N. WITNESSES:

Memorandum of Association



Rules and Regulations of the Society made at the time of Society Registration

Aims and Objectives of the Association

Since inception, the aims and objectives mentioned in PPAI memorandum have been based on a set of core underlying aspirations of advancing the cause of plant protection research and development, disseminate plant protection knowledge and maintain a consistent link among the plant protection workers belonging to diverse organizations in the country. These objectives have undergone revision and modifications to align with advances in research and development, technologies as well as infrastructure growth in areas related to plant protection. Following are the existing objectives of the

- Association: To advance the cause of plant protection by encouraging and promoting plant protection research and education in the country.
- To provide a forum for dissemination of information related to plant protection through a journal.
- To bring about exchange of views and ideas on plant protection by organising lectures, discussion, seminars, symposia etc.
- To maintain a link between plant protection workers in India.
- To involve in any such activity which will be useful for the advancement and adoption of plant protection techniques developed by the scientific community.

Membership

a. **Enrolment of members:** Membership of the Association is open to all persons interested in plant protection, who are above 18 years of age.

b. **Categories of members:**

i **Charter members:** Members joined on or before 15th October, 1972 were designated as charter members.

ii **Ordinary members:** Members joining after 15th October, 1972 are ordinary members.

iii **Honorary Life members:** Individuals who are distinguished scientists and have rendered valuable service to the cause of plant protection may be enrolled as Honorary life members on the recommendation of the Executive Council and ratified by the General Body.

Patrons: As per the original memorandum, the Plant Protection Adviser to the Government of India is its honorary patron. At present, eminent people from all walks of life can be patrons. All applications for membership should be approved by the Executive Council.

Fellows: Five years continuous membership as ordinary members shall entitle them to become fellows of the Association and in the case of Charter members, the period is 3 years.

Admission fee and Subscriptions

1. The annual subscription of members is ₹1500/- payable in advance, and shall become due on 1st April of every year. Members and Institution subscribers shall have to bear the bank commission in respect of outstation cheques.
2. Members whose subscriptions happen to be in arrears for more than an year shall be excluded from the privileges of membership until they have paid their arrears. A list of all such members

shall be submitted by the Secretary to the President at each annual meeting and it shall be read out at the request of any member present in the meeting.

3. Towards the end of the year, the Treasurer shall send a reminder to every member whose subscription is in arrears and if the arrears are of more than two years outstanding on the member's name will be automatically removed from Sssociation Register. Three months' notice shall be given before actually removing the names.
4. Those members who are to pay the arrears of subscription for two and more years and whose names have been struck off the rolls of memberships shall be eligible to readmission provided the admission fee together with two years subscription according to the rates in the current years are paid. Such members will be considered as readmitted.
5. Members paying Rs.8000/- shall be designated as Life Members.
6. Life Members after completion of 10 years, may have to donate the difference of amount to make it on par with the existing rate, which may be paid in two or three instalments. Such donations should not be collected in future again. If any member is not willing to donate the same, he/she will enjoy all the privileges of a member but the journal will not be sent to such members. All the Life Members will receive the journal for 10 years from the date of joining.

General Body

General Body shall consist of all the members of the society including life and Annual Members.

Functions

1. To elect the Executive Council for the management of Association
2. To pass the budget for the ensuing year and approve the expenditure statement of the previous year
3. To approve the report of the activities of the Society
4. To appoint an auditor

Executive Council

1. There shall be an Executive Council, which will be responsible to carry out all affairs of the Association.
2. The Executive Council shall consist of Honorary Patron, the President; Vice President; General Secretary; Assistant Secretary; Treasurer and Eight Councillors. All except the Honorary Patron (Plant Protection Adviser to the Government of India) will be elected by the General Body. The Chief Editor shall be ex-officio member of the Council.
3. The General Secretary, Treasurer and Assistant Secretary shall be elected from the place where the headquarters of the Association is located. All the office bearers shall be elected for a period of two years. Any vacancy occurring within the period of two years shall be treated as interim vacancy.
4. All elections to the offices of the Association shall take place at the Annual General Body meeting. Interim vacancies are filled up by the Executive Council from amongst the members. Voting for the election of office bearers shall be by sending a ballot by post to the General

Secretary who will open them at the General Body meeting. Necessary formalities in connection with the elections (inviting nominations, dispatch of ballot papers etc.) will be completed by the General Secretary, one month in advance of dates prescribed for voting.

5. The General Secretary shall be responsible for recording the proceedings of the meetings of the Association and of the Council and for conducting day to day business of the Association. The Assistant Secretary shall assist the Secretary in all matters connected with the Association, and shall be in charge in the absence of the Secretary. The Treasurer shall keep the account, receive deposits and disburse money on behalf of the Association according to directions of the Council.
6. The funds shall be spent only to the attainment of the objects of the society and no portion thereof shall be paid or transferred directly or indirectly to any of its members through any means.
7. The funds of the association shall be in the name of the Plant Protection Association of India operated jointly by General Secretary and Treasurer and the latter shall make payments endorsed by the secretary in respect of bills not exceeding ₹100/- at a time, Bills for sums in excess of this amount shall be jointly endorsed by the secretary and the president to the extent of ₹1000/- before they will be valid for payment by the treasurer.
8.
 - i) There shall be a reserve fund of the association.
 - ii) The reserve fund shall not be used on recurring items.
 - iii) All the admission fees and 50% of life membership fee received shall form the reserve fund.
 - iv) The executive council by decision of the general body by two thirds majority, can sanction any expenditure out of the reserve fund but not exceeding 50%.
9. All accounts of the Association shall be audited annually by an accountant nominated by the executive council. The treasurer shall submit to the accountant by the end of April each year the accounts of the Association for the previous year for the purpose of auditing. The auditor's statement of accounts shall be circulated to all members before the annual General Body meeting.

Meetings of the Council

- Council meetings shall be held periodically at the discretion of the President or at the request of 1/3 of the members of Council.
- The date and time of holding each meeting of the Council shall be notified by the Secretary to each member of the Council at least 2 weeks before the date on which such meetings are to be held. Non-receipt of such notice by any member of the Council shall in no way invalidate the proceedings of the meeting. One-third of the members of the executive Council shall form a quorum.
- Extraordinary meetings of the Council shall be called by the President on a requisition signed by the members of the Council.

Meetings of the Association

1. Ordinary meetings of the association shall be presided over by the President or in his absence by one of the Vice-Presidents or by one of the members of the council present or by member elected at the meetings.
2. An extraordinary meeting of the association will be called if one third of the ordinary members of the association send a signed requisition to the general secretary for such meeting.

Annual General Body Meeting

1. The date of Annual General Body meeting shall be announced by the Council at least four weeks before the meeting and all notices of motions shall be received by the General Secretary by the end of October. The out-going President will address the meeting. In case the President is unable to attend the Annual General Body meeting, the presidential address will be sent and read at the meeting.
2. The venue of the meetings shall be decided by the Executive Council.
3. The agenda and resolutions for the Annual General Body meeting shall be circulated to the members before the date of the meeting.
4. The reading of technical papers will be fully encouraged at all meetings of the Association. Discussions, purely of plant protection interest could however be held at the Annual General Body meetings.

Withdrawal or Removal of members

1. A member may withdraw from the association by signifying his/her desire to do in a letter Addressed to the Secretary. The Association shall not be liable to return any fee that may have been paid by the member in advance.
2. A member who has withdrawn and whose subscriptions are not in arrears shall automatically recover the privileges of membership without reelection, if he joins the Association within a year of withdrawal and he shall be liable to pay all dues as if he had not withdrawn at all.
3. The General Body shall have the power to remove any member of the Association from its rolls.

Amendment of the Rules

1. No amendment of alteration shall be made in purpose of the Association unless it is voted by 2/3rd members present at a meeting convened for the purpose and confirmed by 2/3rd of the members present at a second special meeting.
2. Proposals for an amendment of these rules may be made either by the council or any member duly seconded. In the latter case, it must reach the secretary in writing in time for circulation to all members of the Association at least one month before extraordinary meeting by the end of October for the annual meeting.
3. The executive council may frame or alter bye-laws for efficient working of the Association not inconsistent with its rules.

Winding Up

In case, the Society/Association had to be woundup, the property and funds of the Society/Association that remain after full satisfaction of the liabilities of the Society/Association will be transferred or paid to some other institution with similar aims and objectives.

Activities of Association

The PPAI is working for diverse community of global scientists that provides credible and beneficial information related to plant protection; advocates and participates in the exchange of knowledge among public, policy makers, and the larger scientific community; and promotes as well as provides opportunities for scientific communication, career preparation and professional development for its members.

1. Publications of PPAI

A. Journal: In 1973, the Association started to publish a scientific journal called the “Indian Journal of Plant Protection” to advance the cause of plant protection. Earlier, it was published half yearly and later 2012 onwards,, four issues of the Journal are being brought out every year in March, June, September and December. So far 50 volumes of the journal have been published. The journal is indexed/ abstracted in Bioscience, Chemical Abstracts, Review of Applied Entomology, Review of Plant Pathology, other CAB abstracts and AGRINDEX of FAO.

Indian Journal of Plant Protection

Focus and Scope

Original research papers on applied aspects of plant protection (insect pests, mites, nematodes, diseases, rodents, weeds and plant protection equipment) are considered for publication. Papers on routine laboratory/ field studies are discouraged. Research articles based on Integrated Pest Management trials conducted for a minimum of two seasons shall be given due weightage. Field data especially on assessment of crop losses, control trials and host plant resistance should be for a minimum of two seasons.

Peer Review Process

- Editor/Associate Editor completes the preliminary screening/editing, which includes checking format (Title, affiliations, present address, abstract, scientific nomenclature, text and reference presentation).
- The referees will be selected based on expertise in one or more areas and should be able to think clearly and logically, and also write a good critique.
- Each paper submitted for publication is sent to one independent referee for peer review. In few cases it will be sent to two independent referees.
- Referees are expected to respond to the editor’s request for advice within a stipulated period (15 days), which is clearly stated by the editor. If the review process is not completed within the stipulated time, reminder will be sent to the referee twice at weekly interval.
- Evaluation sheet is also provided to the referee for seeking advice on all aspects of the article. The referee is advised to give willingness for reviewing immediately. If it is not feasible to

review the manuscript within the stipulated period, the reviewer should express his/her inability to take up the task to the editor. The editor can send the article to another referee without further delay.

- On receipt of comments from the referee, Associate Editor/ Chief Editor takes suitable action and paper is sent to author for modification along with the comments of referee/ Associate Editor/ Chief Editor.
- If the reports of referees disagree with regard to the suitability of the paper for publication, Chief Editor will take final decision independently or in consultation with Associate Editor/ Editorial Board on the rejection of the paper and advice of a second referee is also sought, if required.

Publication Frequency

Four issues of the journal are published every year in March, June, September and December, respectively.

Publication Platform and Open Access Policy

The online version of journal is published in Online Journal Publishing System for Research Journals in Indian Council of Agricultural Research and it is available in <https://epubs.icar.org.in/ejournal/>. This journal provides immediate open access to its content on the principle that making research freely available to the public supports a greater global exchange of knowledge.

B. Books/Manuals/Monographs/Proceedings: Plant Protection Association of India has brought out a number of useful publications from time to time. Advances made in the field of plant protection have been compiled and published for the benefits of different stakeholders.

2. Organisation of National/International conferences/symposia/workshops/brainstorming sessions / lectures by eminent scientists

- PPAI regularly organizes workshops and seminars on various themes related to plant health management. It also conducts meetings of plant protection workers affiliated to ICAR institutes, SAUs, state department of agriculture, private sector agencies and other stakeholders associated with plant protection activities.
- The Society also arranges popular lectures every now and then by eminent scientists/plant protection research workers for the benefit of its members and conducts an Annual General Body Meeting with a technical session.

3. Awards by Plant Protection Association of India

PPAI initiated giving awards to encourage activities and scientists to promote the cause of plant protection in India. Following are the awards instituted and given through the Association.

A. Dodla Raghava Reddy Memorial Award

This award was instituted by Dr D V R Reddy, former President of PPAI and former Principal Scientist and Leader of Virology, International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) in the year 1983 in memory of his late father Shri Dodla Raghava Reddy for outstanding

contributions in the field of Plant Protection. The award is given to Indian citizens once in three years constituting a gold medal and a certificate from interest accrued from a fund of Rs.300,000/- (Rupees three lakhs only).

Individuals who wish to be considered should send FIVE copies of their biodata, indicating precisely the achievements made in Plant Protection to ‘The General Secretary, Plant Protection Association of India, NBPGR-Regional Station, Rajendranagar, Hyderabad- 500 030, Telangana State.

The following are the criteria considered for the award:

- The award is restricted to Indian citizens only.
- Only those who are working in institutes supported by the Indian Government and State governments are eligible. These include ICAR supported research institutes, State government and University Grants Commission supported universities.
- Those who are employed by multinational companies, commercial companies and employed by International Institutes including those supported by Consultative Group for International Agricultural Research, are not eligible.
- Candidate should have published research/review papers in journals with high impact factor of not less than “2”. Those who received meritorious award/s for their research will receive priority.
- Candidates who have made significant contributions towards the functioning and reputation of the Plant Protection Association of India are also eligible.
- Initial screening will be done by the Executive Committee (EC) of the Plant Protection Association of India. A list of potential candidates will be forwarded to Dr D V R Reddy or his nominee and he in consultation with EC a maximum of three candidates will be chosen.
- If there is no consensus of the short listed candidates among the EC members, names of the short listed candidates will be sent to an expert of international reputation in the field of agriculture, preferably in plant protection. The expert will choose the final awardees in consultation with EC represented by the General Secretary and Dr Reddy/ his nominee.

B. Dr Bap Reddy Award for Integrated Pest Management

Dr Bap Reddy, an eminent scientist, who contributed immensely for pest management in India and Asia Pacific for nearly 40 years and retired as FAO representative after serving the United Nations for two years, donated Rs 15000/- over three decades ago for instituting this award. The interest accrued from this amount is given as award once in two years. The award constitutes a plaque, a certificate and cash. This award is given once in two years.

The purpose of the award is to promote the concept of Integrated Pest Management (IPM). Individuals whether solely or collectively engaged in development, research or extensions to promote the concept of IPM in India are eligible for the award. The award shall not be given to the same person twice. Work done in India during the previous five years period is considered for the award.

Individuals who wish to be considered should send five copies of their biodata, indicating precisely the work done and the bearing it has on IPM, to the General Secretary, Plant Protection Association of India, NBPGR- Regional Station, Rajendranagar, Hyderabad-500030, Telangana State.

C. Fellows of Plant Protection Association of India

Fellow will be given to Individuals/ Scientists having completed five years of continuous membership in Plant Protection Association of India.

D. Smt. Kavuri Sarada Memorial Award for “The Best Research Paper” published in the “Indian Journal of Plant Protection”

Candidates are not required to send proposals for this award as the screening committee appointed by ‘Plant Protection Association of India’ chooses two Best Research Papers published in the ‘Indian Journal of Plant Protection’ every year for the award. The award consists of a certificate to each one of the authors of the chosen research paper for “The Best Research Paper”.

E. Best Scientist Awards

- The Best Scientist Awards in Young and Senior Category have been instituted by PPAI starting from the present year (2023) marking the occasion of Golden Jubilee Celebrations of PPAI.
- The awards will be given BIANNUALLY to any individual in recognition of his/her professional contribution in Entomology / Plant Pathology / Nematology and is open to the members of the PPAI who are not more than 60 years of age (for senior category) and 40 years of age (for young category) as on 31st December of the year.
- Award comprises of Citation, Certificate and Cash Prize of Rs.15000/- (Fifteen Thousand only for Senior Category) and Rs 10000/- (Ten thousand for Young Category).

Best Scientist (Young Category) - Guidelines

Name of the Award: Best Scientist Award (Young Category) in Entomology / Plant Pathology / Nematology

Number of Awards (Three): One each in the fields of Entomology / Plant Pathology / Nematology

Purpose of the Award

- To create incentive for research workers in the field of Entomology / Plant Pathology / Nematology
- The award shall be made for significant basic or applied research leading to results of practical importance in the field of plant protection.

Administration of the Award

The PPAI shall have the sole right of selection of the recipient of the award and of the formulation of rules governing such selection.

Eligibility of the Award

The award proposed to be given BIANNUALLY to any individual in recognition of his/her professional contribution in Entomology, Plant Pathology and Nematology, is opened to the members of the PPAI who are not more than 40 years of age as on 31st December of the year.

Best Scientist (Senior Category) - Guidelines

Name of the Award: Best Scientist Award (Senior Category) in Entomology / Plant Pathology / Nematology

Number of Awards (Three): One each in the fields of Entomology / Plant Pathology / Nematology

Purpose of the Award

- To create incentive for research workers in the field in Entomology / Plant Pathology / Nematology.
- The award shall be made for significant basic or applied research leading to results of practical importance in the field of plant protection.

Administration of the Award

The PPAI shall have the sole right of selection of the recipient of the award and of the formulation of rules governing such selection.

Eligibility of the Award

The award proposed to be given BIANNUALLY to any individual in recognition of his/her professional contribution in plant protection, is open to the members of the PPAI and age limit of 40-60 years.

Golden Jubilee of Plant Protection Association of India: 50 years of Progress

During 1972-2022, the PPAI has progressed from a small specialized group of few members with beginnings in the premises of a single central institute to a full-fledged Association of National repute and only one of its kind devoted to the cause of advancement and promotion of plant protection activities in the country. In the last 50 years, PPAI has been managed by a very professional team of scientists and associated research workers aided by skilled supporting group of personnel having experience in financial and administrative management. The association has gone a long way in realizing the original objective of building up a strong network of plant protection workers across the country. Now the association has a strength of over 150 regular members, 425 life members and has 100 Institutional memberships with affiliations to organizations of diverse backgrounds. PPAI has also been functioning strictly as per the society guidelines in conduct of its activities through regularly elected executive councils with members from ICAR research institutes, State Agricultural Universities, Directorate of Plant Protection Quarantine & Storage, National Institute of Plant Health Management as well as research workers associated with private sector agencies working in the field of plant protection.

Executive Council Members from 1972 to 2023

Executive Council - 1972-74		Executive Council - 1975-76	
President	Dr K K Nirula	President	Dr K K Nirula
Vice Presidents	Shri V Lakshminarayana Shri T B Lal	Vice Presidents	Dr K D Paharia Dr L M Patel Dr Hamid Ali
General Secretary	Dr S Hussaine	General Secretary	Dr V Lakshminarayana
Asst. Secretary	Shri G K Mathew	Asst. Secretary	Shri T Rangarajan
Treasurer	Shri P K Menon	Treasurer	Shri S S Lal
Chief Editor	Shri B K Verma	Chief Editor	Shri B K Verma
Editor	Dr S S Hussaine	Editor	Dr S S Hussaine
Councillors	Shri K Rama Rao Shri P N. S. Pradhaw Shri S C Das Shri L M Patel	Councillors	Shri K K Mangal Shri K Rama Rao Shri H R Gauba Dr M C Diwakar Shri Maiku Lal Shri N D Das Shri Laxma Reddy Shri P K Menon

Editorial Board Members	Dr O S. Bindra Dr D S Gupta Dr Zile Singh Dr A R Seshadri Dr Appa Rao Dr R S Dewan Dr T Sankaran Er V C S Sastry	---	---
Chairman and Chief Patron	Dr S N Banerjee, Plant Protection Adviser to GOI	Chairman and Chief Patron	Dr S N Banerjee, Plant Protection Adviser to GOI
Executive Council - 1977-78		Executive Council - 1979-80	
President	Dr K K Nirula	President	Dr N C Joshi
Vice Presidents	Dr N C Joshi Dr Mir Hamid Ali Dr K D Paharia	Vice Presidents	Dr Mir Hamid Ali Dr R Gandhi Dr J A Lowe
General Secretary	Dr Basu Chaudhary	General Secretary	Dr V Lakshminarayana
Asst. Secretary	Shri A P Jain	Asst. Secretary	Shri A P Jain
Treasurer	Shri T Rengarajan	Treasurer	Shri T Rangarajan
Councillors	Shri N D Das Shri B K Verma Dr P R Sarma Shri K K Mangal Dr J Lowe Er V C S Sastry Dr Agnihotrudu Dr Ramesh Gandhi	Councillors	Dr Y K Mathur Dr R L Rajak Dr O N Dhar Dr Bhaskaran Dr Reed Dr Seshadri Er V C S Sastry Dr L M Patel
Chief Editor	Dr V Lakshminarayana	Chief Editor	Dr K K Niralu
Joint Editors	Dr S M A Rizvi Dr V Raghunathan	Joint Editors	Dr V Raghunathan Dr K M Azam
Editorial Board Members	Dr Y L Nene Dr T Sankaran Dr A R Seshadri Dr B H K Rao Dr M D Phatak Dr N C Pant Dr A Appa Rao	Editorial Board Members	Dr Y L Nene Shri B K Verma Shri Maiku Lal Dr. Santhi Sarup Dr M B Kalode Dr B H K Rao Dr V T John Dr B Balasubramanyan
Chairman and Chief Patron	Dr S N Banerjee, Plant Protection Adviser to GOI	Chairman and Chief Patron	Dr K D Paharia, Plant Protection Adviser to GOI

Executive Council - 1981-82		Executive Council - 1983-84	
President	Dr K D Paharia	President	Dr K D Paharia
Vice Presidents	Dr N C Joshi Dr V Lakshminarayana Dr B H Krishnamurthy Rao	Vice Presidents	Dr N C Joshi Dr B H Krishnamurthy Rao Dr R L Rajak
General Secretary	Dr. V. Raghunathan	General Secretary	Dr V Raghunathan
Asst. Secretary	Shri N D Das	Asst. Secretary	Shri N D Das
Treasurer	Dr A Jayaprakash	Treasurer	Dr A Jayaprakash Dr B J Divakar (w.e.f. 14/05/1984)
Councillors	Dr D Bap Reddy Dr J Lowe Dr W Reed Shri K Rama Rao Dr S K Pal Dr P L Premi Shri O N Dhar Dr Gangwane	Councillors	Dr D Bap Reddy Dr D N Srivastava Dr W Reed Dr V Lakshminarayana Dr A D Pawar Dr B V David Shri B K Verma Dr D P Nag Shri K Rama Rao
Chief Editor	Dr K K Niralu	Chief Editor	Dr M Veerabhadra Rao
Joint Editors	Dr V T John Dr T Sankaran Dr S Sarup Dr M B Kalode Dr M Lal	Joint Editors	Dr S Sithanantham Sri B G Naik Sri M Lal Dr S Sarup Dr M B Kalode Dr R D V J Prasada Rao Dr V T John Er V C S Sastry
Editorial Board Members	Dr Y L Nene Dr K A Balasubramanyam Dr M Balasubramanyam Dr R L Rajak	Editorial Board Members	Dr Y L Nene Dr V K Pathak Dr M Balasubramanyam Dr K Krishnaiah Shri T Rangarajan Dr Meera Gupta Dr P S Chandurkar Dr A M K Mohan Rao
Chairman and Chief Patron	Dr K D Paharia, Plant Protection Adviser to GOI	---	----

Executive Council - 1985-86		Executive Council - 1987-88	
President	Dr N C Joshi	President	Dr D Bap Reddy
Vice Presidents	Dr D Bap Reddy Shri B K Verma Dr BV David	Vice Presidents	Dr D V R Reddy Shri B K Verma Dr K Krishnaiah
General Secretary	Shri B Govinda Naik Dr RDVJ Prasada Rao (from 31.01.86)	General Secretary	Dr B J Divakar
Asst. Secretary	Dr RDVJ Prasada Rao Dr Meera Gupta (from 31-1-86)	Asst. Secretary	Dr Meera Gupta
Treasurer	Dr B J Divakar	Treasurer	Dr T Ranga Rajan
Councillors	Shri N D Das Dr A M K Mohan Rao Dr Shanti Swarop Dr S P Singh Dr A S Sindhu Dr Meera Gupta Er V C S Sastry	Councillors	Dr A D Pawar Dr S M A Rizvi Dr T B Gour Dr RDVJ Prasada Rao Dr N V Krishnaiah Dr A S Sindhu Dr S L Taneja
Chief Editor	Dr M Veerabhadra Rao	Chief Editor	Dr M Veerabhadra Rao
Editor	Dr S Sithanantham	Editor	Dr S Sithanatham
Chairman and Chief Patron	Dr R L Rajak, Plant protection Adviser to GOI	Honorary Patron	Dr R L Rajak, Plant Protection Adviser to GOI
Executive Council - 1989-90		Executive Council - 1991-93	
President	Dr S Jayaraj	President	Dr D V R Reddy
Vice Presidents	Dr B V David Dr K. Krishnaiah Dr V Raghunathan	Vice Presidents	Dr K Krishnaiah Dr K P Srivastava Dr B V David
General Secretary	Dr B J Divakar	General Secretary	Dr B J Divakar
Asst. Secretary	Dr N S Ingale	Asst. Secretary	Dr C Pramod Chandra Kumar
Treasurer	Mr T Ranga Rajan	Treasurer	Mr D Chatterjee
Councillors	Dr G T Gujar Dr Meera Gupta Dr K S R K. Murthy Dr A D Pawar Dr S M A Rizvi Dr S P Singh Dr S L Taneja Dr T B Gour	Councillors	Dr K S R K. Murthy Dr Renu Sharma Dr K C Devaraj Urs Dr J R Patel Dr S M A Rizvi Dr Meera Gupta Dr S P Singh Dr A D Pawar

Chief Editor	Dr M Veerabhadra Rao	Chief Editor	Dr M Veerabhadra Rao
Editor	Dr H C Sharma	Editor	Dr H C Sharma
Editorial Board Members	Dr H S Sohi Dr S Nagarajan Dr M V Reddy Dr M S Chary Dr I C Pasalu Dr M D Awasthi Dr B G Naik	Editorial Board Members	Dr S R S Dange Dr T B Gour Dr S K Handa Dr B G Naik Dr T P Sriharan Dr V TSundaramurthy Dr P L Tandon Dr R P Thakur Dr B J Divakar
Honorary Patron	Dr R L Rajak, Plant Protection Adviser to GOI	Honorary Patron	Dr R L Rajak, Plant Protection Adviser to GOI
Executive Council - 1993-95		Executive Council - 1995-97	
President	Dr D V R Reddy	President	Dr D V R Reddy
Vice Presidents	Dr K Krishnaiah Dr P S Chandukar Dr B V David	Vice Presidents	Dr K Krishnaiah Dr P S Chandukar Dr K S R K Murthy
General Secretary	Dr B J Divakar	General Secretary	Dr B J Divakar
Asst. Secretary	Dr C Pramod Chandra Kumar	Asst. Secretary	Dr Renu Sharma
Treasurer	Mr C V Rama Rao	Treasurer	Mr C V Rama Rao
Councillors	Dr Renu Sharma Dr K S R K Murthy Dr Meera Gupta Dr K C Devaraj Urs Dr C S Pawar Dr A M K. Mohan Rao Dr K S Varaprasad Dr Mangal Sain	Councillors	Dr M V Reddy Dr T Ramesh Babu Dr O P Singh Dr G T Gujar Dr B Senapathi Mr O N Dhar Mr Reddy Raju Perumal Dr B S Rathore
Chief Editor	Dr H C Sharma	Chief Editor	Dr T B Gour
Editor	Dr T B. Gour	Editor	Dr C Pramod Chandra Kumar (1995-96) Dr K Chandrasekhara Rao (1996-97)

Editorial Board Members	Dr K Chandrasekhara Rao Dr T Ramesh Babu Dr D S Ajri Dr N Ramakrishnan Dr A N Mukhopadhyay Dr D J Patel Dr Ajay Varadachary Dr B J Divakar	Editorial Board Members	<u>1995-96</u> Dr K C S Rao Dr T Ramesh Babu Dr G K Veeresh Dr D S Ajri Dr N Ramakrishnan Dr AN Mukhopadhyay Dr D J Patel Dr A Varadachary Dr G R S Reddy Dr B J Divakar <u>1996-97</u> Dr John A Wightman Dr T N Raju Dr S B Sharma Dr RDVJ Prasada Rao Dr N V Krishnaiah Dr D D R Reddy Dr C Chattopadhyay Dr B J Divakar
Honorary Patron	Dr R L Rajak, Plant Protection Advisor to GOI	Honorary Patron	Dr V Rahunanthan, Plant Protection Adviser to GOI
Executive Council - 1997-99		Executive Council - 2000-02	
President	Dr K Krishnaiah	President	Dr P S Chandukar
Vice Presidents	Dr P S Chandukar Dr K S R K Murthy Dr B Senapathi	Vice Presidents	Dr K S R K Murthy Mr O N Dhar Dr S M A Rizvi
General Secretary	Dr Renu Sharma	General Secretary	Dr RDVJ Prasada Rao
Asst. Secretary	Mr D Chattopadhyay	Secretary	Mr D Chattopadhyay
Treasurer	Mr C VRama Rao	Treasurer	Mr D Chattopadhyay
Councillors	Dr T Ramesh Babu Dr S M A Rizvi Dr V R. Bhagwat Dr P V Sharma Dr B Padmanaban Dr B K Sahoo Dr B C Shukla Dr T P Rajendran	Councillors	Dr B Senapathi Dr A J Tamhankar Dr S K Gangwar Dr P Sundara Raju Dr P P Shastry Dr S K. Chakrabarty Dr G V Ranga Rao Dr P V Sarma
Chief Editor	Dr T B Gour	Chief Editor	Dr K S Varaprasad

Editor	Dr K Chandrasekhara Rao	Editor	Dr B Sarath Babu
Editorial Board Members	Dr. N. V. Krishnaiah Dr. S. D. Sharma Dr. RDVJ Prasada Rao Dr. D. D. R. Reddy Dr. C. Chattopadhyay Dr. Renu Sharma	Patron	Dr K S R. K. Murthy
Honorary Patron	Dr. R. L. Rajak, Plant Protection Advisor to GOI	Honorary Patron	Dr. V. Raghunathan, Plant Protection Advisor, GOI
Executive Council - 2003-04		Executive Council - 2005-06	
President	Dr. Y. L. Nene	President	Dr. Y L Nene
Vice Presidents	Dr. K. S. R. K. Murthy Dr. H. D. Shirma Dr B. Julius Diwakar	Vice Presidents	Dr. K. S. R. K. Murthy Dr. O. N. Dhar Dr. R. K. Khetrapal
General Secretary	Dr. RDVJ Prasada Rao	General Secretary	Dr. RDVJ Prasada Rao
Asst. Secretary	Dr. V. Markandeya	Asst. Secretary	Dr. C. Pramod Chandra Kumar
Treasurer	Dr. K. Anitha	Treasurer	Dr. S. K. Chakrabarty
Councillors	Dr. R. C. Dani Dr. C. Chattopadhyay Dr. S. K. Gangwar Dr. M. K. Naik Dr. Ashok Krishna Dr. S. K. Chakrabarty Dr. G. V. Ranga Rao Dr. P. V. Sarma	Councillors	Dr. B. K. Mishra Dr. C. Chattopadhyay Dr. S. K. Gangwar Dr. M. K. Naik Dr. A. J. Thamankar Dr. S. K. Chakrabarty Dr. P. R. Yadav Dr. P. V. Sarma Dr. R. K. S. Tiwari Dr. N. V. Krishnaiah Dr. A. Raja Reddy Dr. Asif Tanveer
Chief Editor	Dr. K. S. Varaprasad	Chief Editor	Dr. B. Sarath Babu
Editor	Dr. B. Sarath Babu	Editor	Dr. K. Anitha
Editorial Board Members	Dr. RDVJ Prasad Rao (EX-officio) Dr. K. Muralidharan Dr. R. P. Thakur Dr. Harvir Singh Dr. Gururaj Katti Dr. T. V. K. Singh Dr. Arjun Lal	Editorial Board Members	Dr. RDVJ Prasad Rao (EX-officio) Dr. K. Muralidharan Dr. R. P. Thakur Dr. Harvir Singh Dr. J. S. Prasad Dr. T. Ramesh Babu Dr. Gururaj Katti Dr. T. V. K. Singh Dr. J. R. Felerio Dr. V. Celia Chalam
Honorary Patron	Dr P S Chandurkar	Honorary Patron	Dr. P. S. Chandurkar

Executive Council - 2007-09		Executive Council - 2010-13	
President	Dr. K. S. R. K. Murthy	President	Dr. K. S. Varaprasad
Vice Presidents	Dr. RDVJ Prasada Rao Dr. Harvir Singh Dr. D. J. Patel	Vice Presidents	Dr. R. J. Rabindra Dr. T. Ramesh Babu Dr. C. Chattopadhyay
General Secretary	Dr. S. K. Chakrabarty	General Secretary	Dr. B. Sarath Babu
Asst. Secretary	Dr. Gururaj Katti	Asst. Secretary	Dr. A. Raja Reddy
Treasurer	Dr. Kamala Venkateswaran	Treasurer	Dr. S. K. Chakrabarty
Councillors	Dr. P. K. Sarangi Dr. P. D. Meena Dr. V. B. Nargund Dr. A. Raja Reddy Dr. Suseelendra Desai Dr. Ashish Kumar Tripathi Dr. R. L. N. Murthy Dr. P. S. Chandurkar	Councillors	Dr. P. K. Singh Dr. B. S. Agnihotri Dr. A. K. Tripathi Dr. K. P. Srivastava Mr. G. Ravi Dr. Suryakala Mr. Ashok Jalagam Dr. Asif Tanveer
Chief Editor	Dr. B. Sarath Babu	Chief Editor	Dr. Gururaj Katti
Editor	Dr. Anitha Kodaru	Editor	Dr. G. S. Laha
Editorial Board Members	Dr. RDVJ Prasad Rao (EX-officio) Dr. K. Muralidharan Dr. R. P. Thakur Dr. R. K. Khetrapal Dr. G. V. Ranga Rao Dr. J. S. Prasad Dr. T. V. K. Singh Dr. H. C. Sharma Dr. C. S. Reddy Dr. G. V. Subbaratnam Dr. C. Chattopadhyay	Editorial board members	Dr. H. C. Sharma Dr. S. Ramani Dr. R. K. Khetrapal Dr. K. P. Jayant Dr. S. J. Rehman Dr. C. Chattopadhyay Dr. Suseelendra Desai Dr. Shashi Bhalla Dr. N. Somasekar Dr. S. Ram Singh
Honorary Patron	Dr. K. S. Varaprasad	Honorary Patron	Dr. P. S. Chandurkar Dr. K. S. R. K. Murthy Mr. N. Sukumar

Executive Council - 2014 - 15		Executive Council - 2016-18	
President	Dr K S Varaprasad	President	Dr K S Varaprasad
Vice Presidents	Dr T Ramesh Babu Dr N Sathyanarayana Dr Rajan Sharma	Vice presidents	Dr B Sarath Babu Dr T Srinivasa Rao Dr N Sathyanarayana
General Secretary	Dr B Sarath Babu	General Secretary	Dr Anitha Kodaru
Asst. Secretary	Dr A Raja Reddy	Asst. Secretary	Dr A Raja Reddy
Treasurer	Dr K Rameash	Treasurer	Dr Kamala Venkateswaran
Councillors	Dr S R. Koteswara Rao Dr G Shyam Prasad Dr P C Rath Dr L K Rath Dr S Vennila Dr Baleshwar Singh Dr Alpeshkumar V Khanpara Dr Ajit Singh Dr B L Jat Dr Chandish R Ballal Dr A Muthukumar Dr Asif Tanveer	Councillors	Dr P P Jambhulkar Dr Alpeshkumar V Khanpara Dr Ashish kumar Tripathi Dr Asif Tanweer Dr B K Sahoo Dr K Rameash Dr Kavita Gupta Dr D. P. Kamala Jayanthi Dr R Jagadeeshwar Dr R K Sharma Dr S R Koteswara Rao Dr Y Sridhar
Chief Editor	Dr Anitha Kodaru	Chief Editor	Dr Rajan Sharma
Editor	Dr G Sridevi	Associate Editors	Dr Mukesh Kumar Dhillon Dr Rajeve Rathour Dr Jameel akhtar Dr Cherukuri Srinivasa Rao
Editorial Board Members	Dr B Sarath Babu Dr Josep A J Miret Dr Mathyam Prabhakar Dr Jameel Akhtar Dr Rob Tanner Dr Mohammad Akram Dr V. B. Nargund Dr Hamadttu Abdel Farag Elshafie Dr N. Somasekar Dr P Satya Srinivas	Editorial Board Members	Dr Anitha Kodaru Dr R P Thakur Dr Muthyam Prabhakar Dr Md Akram Dr Hamadttu Dr K Vijaya Lakshmi Dr Dilip K. Ghosh Dr T Uma Maheswari
Honorary Patrons	Dr S N Sushil Dr K S R K Murthy Mr N Sukumar	Honorary Patrons	Dr. A. K. Sinha Dr. K. S. R. K. Murthy Mr. N. Sukumar
Executive Council - 2018-21		Executive Council - 2021-23	

President	Dr B Sarath Babu	President	Dr. B. Sarath babu
Vice presidents	Dr Gururaj Katti Dr T Srinivasa Rao Dr A Raja Reddy	Vice Presidents	Dr. R. Jagadeeshwar Dr. V. Celia Chalam Dr. M. Srinivas Prasad
General Secretary	Dr R Jagadeeshwar	General Secretary	Dr. B. Parameswari
Asst. Secretary	Dr G Shyam Prasad	Asst. Secretary	Dr V Prakasam
Treasurer	Dr Prasanna Holajjer	Treasurer	Dr Bhaskar Bajar
Councillors	Dr P P Jambhulkar Dr Alpeshkumar V Khanpara Dr Asif Tanweer Dr Ashish kumar Tripathi Dr B K Sahoo Dr K Rameash Dr Kavitha Gupta Dr D P Kamala Jayanthi Dr R Jagadeeshwar Dr R K Sharma Dr S R Koteswara Rao Dr Y Sridhar	Councillors	Dr B S Gotyal Dr Alpeshkumar V Khanpara Dr D Sagar Dr K Rameash Dr J Stanley
Chief Editor	Dr G Sridevi	Chief Editor	Dr L Saravanan
Associate Editor	Dr V Ramya	Associate Editors	Dr Kavitha Gupta Dr Prasanna Holajjer
Editorial Board Members	Dr Anitha Kodaru Dr R P Thakur Dr Muthyam Prabhakar Dr Md Akram Dr Hamadttu Abdel Farag Elshafie Dr K Vijaya Lakshmi Dr Dilip K Ghosh Dr T Uma Maheswari	Editorial Board Members	Dr J Alice R P Sujeetha Dr Jameel Aktar Dr Kuldeep S. Jadon Dr J R Falerio Dr Hamadttu Abdel Farag Elshafie Dr P Anandi Dr P Roja Dr Ravindra Chandra Joshi Dr D Sagar
Honorary Patrons	Dr. A. K. Sinha Dr. K. S. R. K. Murthy Mr. N. Sukumar	Honorary Patrons	Dr. A. K. Sinha Dr. K. S. R. K. Murthy Mr. N. Sukumar

PPAI Activities

Publication of National Journal:

Since its inception in 1973, the publication of the journal, Indian Journal of Plant Protection (IJPP) has been one of the foremost and key activities of the association. Four issues of the Journal are published every year in March, June, September and December. So far 50 volumes have been published and the journal has gained repute as being one of the very few national journals catering to the requirements of disseminating knowledge related to plant protection. IJPP is indexed/abstracted in Bioscience, Chemical Abstracts, Review of Applied Entomology, Review of Plant Pathology, other CAB abstracts and AGRINDEX of FAO. The journal has also been accorded the standard and widely accepted NAAS rating enhancing its acclaim in recognizing the importance of publishing plant protection related research findings in the interest of different stakeholders and with final benefits to farmers.

Organization of Conferences/Workshops/Symposia/Brain storming sessions

The Association has organized the following conference/symposia/workshop so far:

- ▶ 1986 - National Seminar on Plant Protection in Field Crops-Hyderabad
- ▶ 1988 - National Workshop on Plant Protection in Horticultural crops-Hyderabad
- ▶ 1992 - National Seminar on Changing Scenario in Pest and Pest Management-Hyderabad
- ▶ 1994 - National Workshop on use of Pheromones in Integrated Pest Management-Hyderabad
- ▶ 1995 - National Symposium on Integrated Pest Management and Environment-Hyderabad
- ▶ 1997 - National Seminar on Plant Protection Towards Sustainability-Hyderabad
- ▶ 2007- National Conference on Organic Waste Utilization and Eco Friendly Technologies for Crop Protection-Hyderabad
- ▶ 2012 - International Conference on Plant Health Management for Food Security, Hyderabad
- ▶ 2016 - National Conference on Plant Health Management, Tirupathi
- ▶ 2018 - Brain Storming Session on Emerging Plant Protection Technologies: Opportunities and Challenges, Hyderabad
- ▶ 2019 - Associated in organizing “XIX International Plant Protection Congress” (IPPC2019), Hyderabad
- ▶ 2019 - Red Palm Weevil Symposium: - Outsmarting the Red Palm weevil: A Global Challenge during XIX International Plant Protection Congress (IPPC2019), Hyderabad
- ▶ 2023- Brain Storming Session on Integrated Pest Management of Invasive Black Thrips (*Thrips parvispinus*) in Chilli in Telangana and Andhra Pradesh, Hyderabad

Books and Proceedings of Conferences

PPAI has released several important publications during these national and international group events. It has also published brochures, booklets and related material for the benefit of students, researchers, state department personnel and farmers:

- ▶ **Plant Protection in Field Crops: Proceedings of National Seminar on Plant Protection in Field Crops**, held at CPPTI, Hyderabad from January 29-31, 1986; Plant Protection Association of India, 1986; 91p.
- ▶ **Pests and Pest Management in India: The Changing Scenario** ; Lead papers of the National Seminar on Changing Scenario in Pests and Pest Management in India ; 31 Jan - 1 Feb 1992, held at CPPTI, Hyderabad, India. Plant Protection Association of India, 1993; 312 p.
- ▶ **Pheromones in Integrated Pest Management**: Proceedings of the National Workshop, 1994; Plant Protection Association of India, Hyderabad.
- ▶ **Plant Protection and Environment**: Proceedings of the National Symposium, 1995, Plant Protection Association of India, Hyderabad
- ▶ **Plant Protection Towards Sustainability – A Special issue of IJPP**: Proceedings of the National Symposium, 1997; Plant Protection Association of India, Hyderabad
- ▶ **Resources Management in Plant Protection**: Papers presented at the National Seminar on Resources Management in Plant Protection during Twenty First Century, held at Hyderabad during 14-15 November 2002; Plant Protection Association of India, 2002; Two Volumes.
- ▶ **Extended Summaries** of the National Conference on Organic Waste Utilization and Eco-friendly Technologies for Crop Protection. 2007; Plant Protection Association of India, Hyderabad.
- ▶ **Forty Years of Plant Protection in India**
- ▶ **Plant Health Management for Food Security – Issues and Approaches**: Proceedings of the International Conference on Plant Health Management for Food Security 2012- Plant Protection Association of India, NBPGR Hyderabad. Astraa Publications. 2016.

Awards by Plant Protection Association of India

On its progressive path with aforementioned objectives in focus, the Association has instituted several awards and recognized the meritorious services and research in plant protection. In the last five decades, the following awards and recognitions were conferred encouraging meritorious scientists, students and academicians from time to time based on the guidelines set for each of those awards.

Recipients of Awards given by Plant Protection Association of India

1. Dodla Raghava Reddy Memorial Gold Medal Award

Year	Name
1997	Dr A Appa Rao
2006-08	Dr C D Mayee
2008-10	Dr K S Varaprasad
2010-12	Dr T Ramesh Babu
2014-16	Dr P Anand Kumar
2016-18	Dr A G Sreenivas

2. Dr Bap Reddy Award for Integrated Pest Management

Year	Name	Year	Name
1986-1988	Dr K Krishnaiah	2002-2004	Dr B Srinivasulu
1988-1990	Dr A D Pawar	2006-2008	Dr N P Eswar Reddy
1990-1992	Dr S Uthamasamy	2008-2010	Dr S Suresh
1992-1994	Dr T M Manjunath	2010-2012	Dr Y G Prasad
1994-1996	Dr V Muniappa	2014-2016	Dr U Sreedhar
1998-2000	Dr H C Sharma	2016-2018	Dr S Vennila
2000-2002	Dr I C Pasalu and Team		

3. Fellows of Plant Protection Association of India

Before 2003: Dr D Bap Reddy, Dr N C Joshi, Dr B V David, Dr K Krishnaiah, Dr V C S Sastry, Dr B J Divakar, Dr H C Sharma, Dr L V Gangwane, Dr Viswanath, Dr S M A Rizvi, Dr S S Lateef, Dr O P Singh, Dr A D Pawar, Dr K P Srivastava, Dr D J Patel, Dr S S Misra, Dr A R Solayappan, Dr K Natarajan, Dr A K Garg, Dr Allahnoor, Dr S K Gangwar, Dr S K Srivastava, Dr G Arjanan, Dr M Balasubramanian, Dr D Kumaresan, Dr. M. Veerabhadra Rao, Dr G Maruthi Ram, Dr K C Bhagat, Dr V K Kalra, Dr K S R K Murthy, Dr Shashi Verma, Dr K C D Urs, Dr R C Joshi, Dr S R Das, Dr G Balasubramanian, Dr H P Patnaik, Dr S Kumarasamy, Dr Rambabu Gaur, Dr P V Sarma, Dr G T Gujar, Dr T V K Singh, Dr V Raghunathan, Dr S P Singh, Dr T B Gour, Dr K C S Rao, Dr H R Patel, Dr Raghunatha Rao, Dr Sanjay Kumar, Dr I V Dhruj, Dr R T Gahukar, Dr B Senapathi, Dr G Ramaprasad, Dr S V Damdhere, Dr N C Misra, Dr R P Thakur, Dr S B Sharma, Dr Renu Sharma, Dr Meera Gupta, Dr Amrit Phokela, Dr S Jayaraj, Dr K Abbaiah, Dr B K Sontakke, Dr P S Chandukar, Dr N Venugopal Rao, Dr G V Subbarathnam, Dr Suresh Pande, Dr S K Pal, Dr Mangal Sain, Dr S P Sharma, Dr S S Bhardwaj, Dr C P C Kumar, Dr S Uthamasamy, Dr B Padmanabhan, Dr M P Singh, Dr P R M Rao, Dr T M Manjunath, Dr R D V J Prasada Rao, Dr T Ramesh Babu, Dr P R Misra, Dr Ravi Prakash Maurya, Dr A J Tamhankar, Dr V R Bhagwat, Dr M K Naik, Dr Prakash P Shastry, Mr Satish Parsai, Dr Madhuban Gopal, Mr J Kannaiyan, Dr M Mani, Dr Chirantan Chattopadhyay, Dr P V Krishnayya, Dr D Jagadishwar Reddy, Dr K K Pandey, Prof S V Sarode, Prof A Regupathy, Dr B K Mishra, Dr Ashok Krishna, Dr H P Mishra, Dr (Mrs) Saxena and Mr V Ambethgar

2003: Dr U Sreedhar, Dr Prem Raj Gupta, Dr Kanchan Baral, Dr B K Sahoo and Dr Rajgopal

2005: Dr M A Raoof, Dr V B Nargund, Dr Gururaj Katti, Dr Asif Tanweer and Dr P Nagaraja Rao

2006: Dr S Desai, Dr Subrata Satpathy, Dr P Sreerama Kumar, Dr Ladu Kishore Rath and Dr Anitha Kodaru

2010: Dr A Rajareddy, Dr Satya Vir, Dr V Ramesh Babu, Dr RW Alexander Jesudasan, Dr S Manickavasagam, Dr V Jayalakshmi, Dr Yashoda R Hegde, Dr Pankaj Sharma, Prof Ram Singh, Dr V Selvanarayanan, Dr Rajesh Pratap Singh, Dr S Jeyarani, Dr R Velazhahan, Dr PS Vimaladevi, Dr Mukesh K Dhillon, Dr J Jayakumar, Dr Abhishek Shukla and Dr T V K Singh

2012: Dr T S K Patro, Dr P C Rath, Dr P D Meena, Dr Kavita Gupta, Dr S R Pandravada, Dr B Sree laxmi, Dr T Saravanan, Dr G Shyam Prasad, Dr P G Padmaja, Dr Chandish R Ballal, Dr G M V Prasad, Dr Dewa Ram Bajya, Dr S Ramakrishnan, Dr N Somasekhar, Dr Arjun Lal, Dr A P Padma Kumari, Dr M Prabhakar, Dr P K Behera, Dr Sashi Bhalla and Dr V Kamala

2016: Dr V Lakshminarayanamma, Dr Ch Padmavathi, Dr Bharati N Bhat, Dr K Shankarganesh, Dr M Rajasri, Dr R Jagadeeshwar, Dr G Sridevi, Dr K Karthikeyal, Dr M Visalakshmi, Dr B Bhavani, Dr Ashish Kumar Tripathi, Dr Aravindnath Singh, Dr D Karthikeyan, Dr T V Prasad and Dr K Rameash

4. Smt. Kavuri Sarada Memorial Award for “The Best Research Paper” published in the “Indian Journal of Plant Protection”

Year	Names of Authors
1988	Dr K Abbaiah and Dr M Sugunakara Reddy
1989	Dr D V Singh, Dr P Arora, Dr K D Srivastava, Dr S Nagarajan and Dr R Agarwal
1992-93	Dr M Veerabhadra Rao
1993-94	Dr C Pramod Chandra Kumar
1994-95	Dr B J Divakar, Dr P V Sharma, Dr V Raghunathan, Dr B V David, Dr G R S Reddy and Sri S V Swamy
1995-96	Dr P P Shastry
1996-97	Dr T P Sriharan
2001	Dr V Markandeya, Dr V Vasu, Dr PS Chandurkar, Dr T Rangarajan and Dr B J Divakar
2002	Dr M Mani and Dr A Krishnamoorthy
2004	Dr T K S Latha
2007	Satya Vir (2007). Neem Genetic Diversity in India and its Use as Biopesticide and Biofertilizer. <i>Indian Journal of Plant Protection</i> , Vol. 35.
2008	V Ramesh Babu (2008). F2 Screen Estimation of Alleles Population of Diamondback Moth (<i>Plutella xylostella</i> Linn.). <i>Indian Journal of Plant Protection</i> , Vol. 36.
2010	V Manoj Kumar (2010). Management economics of foot and stem rot of mesta incited by <i>Phytophthora parasitica</i> . <i>Indian Journal of Plant Protection</i> , Vol. 38.
2011	N Ramakrishnan (2011): Standardization of X-Ray Radiography Methodology for the detection of hidden infestation in cereals. <i>Indian Journal of Plant Protection</i> , Vol. 39.
2012	Ratna Bhimineni
2013	P A Ahila Devi & V Prakasam (2013). RAPD-based Genetic Variation among <i>Colletotrichum</i> Isolates causing Chilli Anthracnose. <i>Indian Journal of Plant Protection</i> , 41(3): 244-248.
2014	A Kandan, J Akhtar, B Singh, U Dev, R Goley, D Chand, A Roy, S Rajkumar, and P C Agarwal (2014). Genetic diversity analysis of <i>Alternaria alternata</i> isolates infecting different crops using URP and ISSR markers. <i>Indian Journal of Plant Protection</i> 42 (3): 229-236.

2015	D Balakrishna, K Srinivasa Babu, B Venkatesh Bhat, R Vinod, M Sreedhar, G Shyamprasad, D B Pawar, Shekharappa, M O Mohammed Ilyas and J V Patil (2015). Improved shoot fly resistant sources by gamma irradiation induced mutation in sorghum, <i>Indian Journal of Plant Protection</i> . Vol 43 (4).
2016	K Susheela and N Sathyanarayana (2016). Weed risk assessment of <i>Ambrosia psilostachya</i> for its invasiveness and potential endangered areas in India. <i>Indian Journal of Plant Protection</i> . Vol 44 (1).

5. Dr R D V J Prasada Rao Award

2012: Dr V Celia Chalam
2016: Dr Kavita Gupta

PPAI Golden Jubilee Awards - ICPHM 2023 [15-18 November 2023]

1. Dodla Raghava Reddy Memorial Award

1. Dr M Srinivasa Rao (2018-2020)
2. Dr Basava Prabhu Patil (2020-2022)

2. Dr Bap Reddy Award for Integrated Pest Management

1. Dr Jameel Akthar (2018-2020)
2. Dr NBV Chalapathi Rao (2020-2022)

3. Fellows of PPAI (2017-2023)

1. Dr Srinivas Parimi, Regional Resistance Management Lead APAC, Bayer Crop Science, India.
2. Dr C Alice Retna Packia Sujeetha, Director - Plant Biosecurity, National Institute of Plant Health Management, Hyderabad.
3. Dr M Srinivas Prasad, Principal Scientist & Head, ICAR-Indian Institute of Rice Research, Hyderabad.
4. Dr Suresh Kumar Khinchi, Associate Professor, College of Agriculture, Agriculture University, Jodhpur.
5. Dr A Muthu Kumar, Associate Professor, Faculty of Agriculture, Annamalai University, Chidambaram.

6.	Dr Jameel Akhtar, Principal Scientist, ICAR-National Bureau of Plant Genetic Resources, New Delhi.
7.	Dr M Alagar, Assistant Professor, Tamil Nadu Agricultural University, Coimbatore.
8.	Dr K P Manju, Assistant Professor, Krishi Vigyan Kendra, Kannur, Kerala Agricultural University
9.	Dr V Prakasam, Senior Scientist, ICAR-Indian Institute of Rice Research, Hyderabad.
10.	Dr R Maruthadurai, Senior Scientist, ICAR-Central Coastal Agricultural Research Institute, Goa.
11.	Dr Pardeep Kumar, Scientist (Senior Scale), ICAR-National Bureau of Plant Genetic Resources, New Delhi
12.	Dr D Ladhalakshmi, Senior Scientist, ICAR-Indian Institute of Rice Research, Hyderabad.
13.	Dr Prasanna Holajjer, Senior Scientist, ICAR-National Bureau of Plant Genetic Resources Regional Station, Hyderabad.
14.	Dr Veghra Durga Prasad Rao Nimmakayala, Ex-consultant, United Nations Environment Programme
15.	Dr L Saravanan, Senior Scientist, ICAR-National Bureau of Plant Genetic Resources Regional Station, Hyderabad
16.	Dr M Punithavalli, Senior Scientist, ICAR-Sugarcane Breeding Institute, Coimbatore
17.	Dr B S Gotyal, Senior Scientist, ICAR-National Bureau of Agricultural Insect Resources, Bengaluru.
18.	Dr K Selvaraj, Senior Scientist, ICAR-National Bureau of Agricultural Insect Resources, Bengaluru,
19.	Dr Kuldeep Singh Jadon, ICAR-Central Arid Zone Research Institute, Jodhpur.
20.	Dr P Duraimurugan, ICAR-Indian Institute of Oilseeds Research, Hyderabad.
21.	Dr Issai Aruna Sri, Scientist, Professor Jayashankar Telangana State Agricultural University, Hyderabad.
22.	Dr V Jhansi Lakshmi, Principal Scientist, ICAR-Indian Institute of Rice Research, Hyderabad.
23.	Dr Kiran Babu Talluri, Scientist, Professor Jayashankar Telangana State Agricultural University, Hyderabad.
24.	Dr Satish Kumar Sain, Principal Scientist, ICAR-Central Institute for Cotton Research, Regional Station, Sirsa

25. Dr SVS Gopala Swamy, Principal Scientist & Vice Principal, Polytechnic of Agriculture & Polytechnic of Agriculture Engineering, Anakapalle.
26. Dr Girish Anantrao Gunjotikar, Deputy Director (Plant Pathology), National Institute of Plant Health Management, Hyderabad
27. Dr K Vemana, Principal Scientist, AICRP-Ground nut, Agricultural Research Station, Kadiri.
28. Dr MH Kodandaram, ICAR-Indian Institute of Pulse Research, Regional Research station, Dharwad.
29. Dr Bonthagorla Subbarayudu, Principal Scientist, ICAR-Indian Institute of Millets Research, Hyderabad
30. Dr J Gulsar Banu, Principal Scientist, ICAR-Central Institute for Cotton Research, Regional Station, Coimbatore.
31. Dr M Visalakshi, Principal Scientist, AICRP on Biological Control, RARS, Anakapalle.
32. Dr D Krishnaveni, Principal Scientist, ICAR-Indian Institute of Rice Research, Hyderabad.
33. Dr S Subramanian Principal Scientist, ICAR-Indian Agricultural Research Institute, New Delhi.
34. Dr MK Jyosthna, Associate Professor, SV Agricultural College, Tirupathi.
35. Dr D Anitha Kumari Principal Scientist, Sri Konda Laxman Telangana State Horticultural University, Hyderabad.

4. Smt. Kavuri Sarada Memorial Award for “The Best Research Paper”

ICPHM 2023 Organizing Committee selected the following research articles for **Smt. Kavuri Sarada Memorial Award for “The Best Research Paper”**, which were published in the Indian Journal of Plant Protection during the year 2019-2022

Year, Volume, Issue no. Page nos.	Title of the paper	Authors, Affiliation, and email for correspondence
2019, Vol. 47 No. 1&2, (21-28)	Efficacy of indigenous strains of entomopathogenic nematodes, <i>Steinernema feltiae</i> and <i>Heterorhabditis bacteriophora</i> against the white grub, <i>Brahmina coriacea</i>	Authors: Isha Sharma, Mohinder Singh and P L Sharma Affiliation: Department of Entomology, Dr. Y. S. Parmar University of Horticulture and Forestry, Nauni, Solan 173230, Himachal Pradesh, India Email: sharmaisha0590@gmail.com

2019, Vol. 47 No. 3 & 4, (154-163)	Marker assisted introgression of broad spectrum blast resistance gene Pi-2 into an elite rice cultivar, Samba Mahsuri	Authors: S Vijay Kumar, M Srinivas Prasad, R Rambabu, B Bhaskar ¹ , R M Sundaram, V Prakasam, D Ladhakshmi, G S Laha and M Sheshu Madhav Affiliation: ICAR-Indian Institute of Rice Research, Rajendranagar, Hyderabad 500 030, Telangana State, India ¹ S V Agricultural College, Acharya N G Ranga Agricultural University, Tirupathi 517502, Andhra Pradesh, India Email: data.msprasad@gmail.com
2020, Vol. 48 No. 3, (190- 196)	Ecological engineering in gourds and melons (Cucurbitaceae) for pest management and conservation of beneficial insects	Authors: E Sree Latha*, S Jesurajan and Ch. Sreenivasa Rao Affiliation: National Institute of Plant Health Management, Rajendranagar, Hyderabad - 500030, Telangana State, India. Email: sreelatha.e@gov.in
2020, Vol. 48 No. 4, (296- 300)	Design and development of a mono wheel operated sprayer cum weeder	Authors: Shambhu Singh, A K Dave, D Padhee and Aman Affiliation: Department of Farm Machinery and Power Engineering, IGKV, Raipur 492001, Chhattisgarh state, India Email: amandhaniya0003@gmail.com
2021, Vol. 49 No. 1, (40-53)	Isolation, characterization of endophytic bacteria and their biocontrol potential against early blight of tomato	Authors: B Sai Sushma ¹ , B Vidya Sagar ² , S Triveni ³ , G Uma Devi ⁴ Affiliation: ^{1, 2, 4} Department of Plant Pathology and ³ Department of Agricultural Microbiology and Bioenergy, Professor Jayashankar Telangana State Agricultural University (PJTS AU), Rajendranagar, Hyderabad 500 030, India. Email: boggarapusaisushma@gmail.com
2021, Vol. 49 No. 4, 2021 (227-232)	Increasing distribution, biology and population dynamics of invasive woolly whitefly, <i>Aleurothrixus floccosus</i> on guava	Authors: K Selvaraj and B V Sumalatha Affiliation: Division of Germplasm Conservation and Utilization, ICAR- National Bureau of Agricultural Insect Resources, Hebbal, Bengaluru 560 024, Karnataka. Email: k.selvaraj@icar.gov.in

2022, Vol. 50 No. 1, (51-55)	Prediction of potential geographic distribution of exotic nematode, <i>Aphelenchoides fragariae</i> in India based on MaxEnt ecological niche modelling	Authors: Prasanna Holajjer*, Bharat H Gawade ¹ , Z Khan ¹ and N Sivaraj Affiliation: ICAR-National Bureau of Plant Genetic Resources, Regional Station, Hyderabad-500030, Telangana State, India 1 ICAR- National Bureau of Plant Genetic Resources, New Delhi- 110012, India. Email: prasannaiari@gmail.com
2022, Vol. 50 No. 4, (115-119)	Mega-field demonstration of management of pink bollworm, <i>Pectinophora gossypiella</i> in rain-fed cotton using mating disruption technology	Authors: C D Mayee ^{1,2*} , B Chaudhary ² , R Panchbhair ¹ , A R Annepu ³ and R D Kapur ³ Affiliation: 1 Agrovision Foundation, Nagpur 2 South Asia Biotechnology Center, Jodhpur-New Delhi 3 PI Industries, Gurugram, Haryana E mail: mayeecharu@gmail.com

PPAI Awards Instituted in 2023

Best Scientist (Senior Category) Entomology

1. Dr Shravan M Haldhar
2. Dr K Selvaraj

Best Scientist (Senior Category) Plant Pathology

Dr V Bhuvaneswari

Best Scientist (Senior Category) Nematology

Dr N Somasekhar

Best Scientist (Young Category) Entomology

Dr Gadratagi Basana Gowda

Best Scientist (Young Category) Plant Pathology

Dr K Sakthivel

Best Scientist (Young Category) Nematology

Dr Satish N Chavan

Best Scientist Digital Agriculture

Dr Srikanth Rupavathara

ICPHM 2023- PPAI Recognition Awards for the Meritorious Services Rendered During 1972-2022

1. Life time Contribution Award

S.No.	Name	Designation
1	Dr Dodla V Raghava Reddy	Former President, PPAI and International Scientist ICRISAT, Hyderabad, India.
2	Dr V Raghunathan	Former Plant Protection Advisor to Government of India, Directorate of Plant Protection, Quarantine & Storage, Ministry of Agriculture, Faridabad
3	Dr P S Chandurkar	Co-Chairman of Scientific Advisory Board of FARMER & Former Plant Protection Advisor to Govt of India, Directorate of Plant Protection, Quarantine & Storage, Ministry of Agriculture, Faridabad
4	Dr K S Varaprasad	Project Manager (USDA-SPS PROJECT), Former Officer-In Charge, ICAR-NBPGR RS, Hyderabad & Former Director, ICAR-IIOR, Hyderabad, India
5	Dr K Krishnaiah	Former President PPAI & Former Director ICAR – IIRR, Hyderabad, India
6	Dr KSRK Murthy	Former President PPAI, Hyderabad, India
7	Dr B Sarath Babu	Former Officer-In Charge, ICAR-NBPGR RS, Hyderabad & President PPAI, Hyderabad, India

2. Outstanding Contribution Award

S.No.	Name	Designation
1	Dr M Veera Bhadra Rao	Former Microbiologist, CPPTI (Present NIPHM), Hyderabad and Former Chief Editor, PPAI
2	Dr B Julius Divakar	Former Entomologist, CPPTI (Present NIPHM), Hyderabad and Former General Secretary, PPAI
3	Dr T B Gour	Former Principal Scientist, ANGRAU
4	Dr Rajan Sharma	Cluster Leader- Crop Protection and Seed Health, ICRISAT, Hyderabad, India
5	Dr T Ramesh Babu	Former Dean of Agriculture, ANGRAU & Dean, School of Agriculture and Food Technology, Vignan University Andhra Pradesh, India
6	Dr Anitha Kodaru	Former Officer-In-Charge and Principal Scientist, ICAR-NBPGR, Regional Station, Rajendranagar, Hyderabad, India

3. PPAI - Golden Jubilee Recognition Award

S.No.	Name	Designation
1	Dr B Govinda Naik	Former General Secretary, PPAI
2	Dr Renu Sharma	Former Nematologist, CPPTI (Present NIPHM), Hyderabad
3	Dr S K Chakrabarty	Former Officer In-Charge & Principal Scientist, ICAR-NBPGR RS, Rajendranagar, Hyderabad, India
4	Dr R Jagadeeshwar	Vice President, PPAI & Former Director of Research, PJTSAU, Rajendranagar, Hyderabad, India
5	Dr B Parameswari	General Secretary – PPAI and Senior Scientist, ICAR-NBPGR, RS Hyderabad, India.
6	Dr R K Khetarpal	Former Head, DPQ, ICAR-NBPGR, New Delhi & Executive Secretary, APAARI, Bangkok
7	Dr Harvir Singh	Former Principal Scientist, ICAR-IIR, Hyderabad, India
8	Dr Gururaj Katti	Former Principal Scientist & Head, Crop Protection, ICAR-IIRR, Hyderabad, India
9	Dr A Raja Reddy	Dean of Agriculture School of Agricultural Sciences Mallareddy University, Hyderabad, India
10	Dr V Celia Chalam	Vice President -PPAI and Head DPQ, ICAR-NBPGR New Delhi, India
11	Dr M Srinivas Prasad	Vice President -PPAI and Principal Scientist, ICAR-IIRR, Hyderabad, India.
12	Dr H C Sharma	Former Vice Chancellor – Dr YS Parmar University of Horticulture & Forestry, Himachal Pradesh & Former Principal Scientist, ICRISAT, Hyderabad, India
13	Dr L Saravanan	Chief Editor, PPAI and Senior Scientist, ICAR-NBPGR, RS Hyderabad, India.
14	Dr K M Azam	Former Deputy Director, CPPTI, Hyderabad and Former University Head (Entomology), APAU
15	Dr S Sithanantham	Managing Director, International Centre of Insect Physiology and Ecology & Plant Health, Hyderabad
16	Dr C Pramod Chandra Kumar	Chief Editor, PPAI and Dean student's welfare, PJTSAU, Hyderabad, India.
17	Dr G Sreedevi	Principal Scientist & Head, AINP on Pesticide Residues, PJTSAU, Rajendranagar, Hyderabad
18	Dr Prasanna Holajjer	Associate Editor, PPAI & Senior Scientist ICAR- NBPGR, RS Hyderabad, India.
19	Dr Bhaskar Bajar	Treasurer -PPAI and Scientist, ICAR- NBPGR, RS Hyderabad, India.

4. PPAI Award of Distinction

S.No.	Name	Designation
1	Dr S N Puri	Former Vice-Chancellor, Central Agricultural University, Imphal, 2004-2014, Director, National Centre for Integrated Pest Management, New Delhi, 1995-99; Vice-Chancellor, Konkan Krishi Vidyapeeth, Dapoli, 1999-2000; Vice-Chancellor, Mahatma Phule Krishi Vidyapeeth, Rahuri, 1999-2004
2	Dr C D Mayee	President of South Asia Biotechnology Centre (SABC) & Former Chairman of Indian Agricultural Scientists Recruitment Board, New Delhi
3	Dr Chelliah	Former Director of Research, TNAU, Coimbatore
4	Dr Anupam Varma	INSA Emeritus Scientist & Former National Professor, IARI, New Delhi
5	Dr A K Dhawan	President, Indian Ecological Society, Punjab Agricultural University, Ludhiana
6	Dr A N Mukhopadhyay	Former Vice-Chancellor of Assam Agricultural University
7	Dr Srikant Kulkarni	Former Professor & Head, Department of Plant Pathology, UAS, Dharwad
8	Dr J S Prasad	Former Principal Scientist and Head Crop Protection, IIRR, Hyderabad
9	Dr H S Gaur	Vice Chancellor at SVPUAT, Meerut & Former Dean, ICAR-IARI, New Delhi

5. PPAI Golden Jubilee Special Award

S.No.	Name	Designation
1	Dr S N Sushil	Director of ICAR-NBAIR, Bengaluru & Former Plant Protection Advisor, Government of India, Directorate of Plant Protection, Quarantine & Storage, Ministry of Agriculture, Faridabad
2	Dr J P Singh	Plant Protection Adviser, Government of India, Directorate of Plant Protection, Quarantine & Storage, Ministry of Agriculture, Faridabad

6. PPAI Posthumous Award

S.No.	Name	Designation
1	Mr Vinod Kumar Samanthula	Founder & Former CEO of Senseacre Labs

Plant Protection Association of India's New Futuristic Journey

Plant Protection Association of India (PPAI) is embarking on a new futuristic journey. Our vision is to serve plant protection communities including businesses, research and farmers of India in pursuit of providing plant health solutions. Enabling this vision, we look to work with public and private sector on critical plant health management topics.

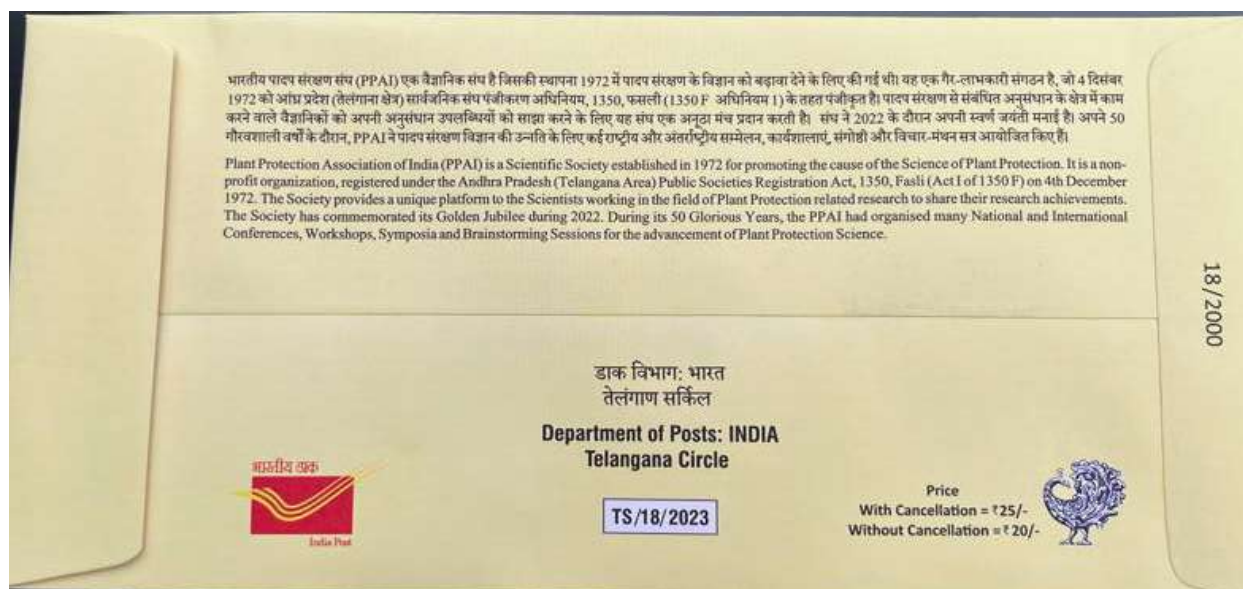
The society under the present Executive Council dreams of expanding its activities to serve the stakeholders viz., researchers, students, the industry, and farmers. The new mandate will be to turn its focus as a leading platform of understanding between the policy bodies of the government and the stakeholders. The society intends to expand its human resources with a panel of scientists drawn from across the country for enhancing the activities like training in plant health management and dealing with farmer friendly new technologies in Digital Agriculture and Farm Mechanization.

PPAI would like to enter into MOUs with the ICAR institutes, the State Agricultural Universities, the industry, the industry connected NGO associations like Federation of Seed Industry of India, BioAgri Input Producers Association, Crocare, Croplife etc as well as some farmer based NGOs.

Therefore, it is quite ambitious to expand its infrastructure, farmer oriented beneficial programmes, human resources and fund raising to meet its futuristic journey successfully.

To better enable this and amplify our efforts we solicit your participation, support, coordination and active contributions.

ICPHM 2023 - Release of Special Postal Cover





పీపీఐఐ గోల్డెన్ జూబ్లీ వేడుకల సందర్భంగా తపాల శాఖ
ఆధ్వర్యంలో ప్రత్యేక పోస్టల్ కవర్ ను విడుదల చేస్తున్న హైదరాబాద్
రీజియన్ పోస్ట్ మాస్టర్ జనరల్ టీఎం శ్రీలత తదితరులు

పీపీఐఐ గోల్డెన్ జూబ్లీ వేడుకల సందర్భంగా ప్రత్యేక పోస్టల్ కవర్ విడుదల

బుద్వేల్, నవంబర్ 10 (ప్రభాత ఉదయం): ప్లాంట్ ప్రొటెక్షన్ అసోసియేషన్ ఆఫ్ ఇండియా స్వర్ణోత్సవాల సందర్భంగా ప్రత్యేక పోస్టల్ కవర్ ను శుక్రవారం విడుదల చేశారు. రాజేంద్రనగర్ పోస్ట్ ఆఫీస్ లో జరిగిన కార్యక్రమంలో హైదరాబాద్ రీజియన్ పోస్ట్ మాస్టర్ జనరల్ టీ.ఎం. శ్రీలత పోస్టల్ కవర్ ను విడుదల చేశారు. ఈ కార్యక్రమంలో ఎన్ఐ పీహెచ్ఎమ్ డైరెక్టర్ జనరల్ డాక్టర్ సాగర్ హనుమాన్ సింగ్, ఎన్ బీ పి జి ఆర్ మాజీ సంచాలకులు డాక్టర్ బి. శరత్ బాబు, డాక్టర్ ఆర్.జగదీశ్వర్, డాక్టర్ జిల్లా సత్యనారాయణ తదితరులు పాల్గొన్నారు. 1972 లో ఏర్పాటైన ప్లాంట్ ప్రొటెక్షన్ అసోసియేషన్ ఆఫ్ ఇండియా 50 సంవత్సరాలు పాటు మొక్కల ఆరోగ్యం కోసం చేస్తున్న కృషి ని గుర్తు చేసుకున్నారు.



India Ka Pranam Har Kisan Ke Naam



For more information call our toll free no. : 1800 102 1022



Dhanuka Agritech Limited

Global Gateway Towers, Near Guru Dronacharya Metro Station,
MG Road, Gurugram -122002, Haryana, Tel. : +91-124-434 5000,
E-mail : headoffice@dhanuka.com, Website : www.dhanuka.com

Log on to: www.dhanuka.com or follow us on:    



Dr. P.V. Ravikumar
CHAIRMAN

Dhana
Crop Sciences Limited

SERVING ALL KINDS OF SPECIALITY MICRONUTRIENTS
INSECTICIDES FUNGICIDES, HERBICIDES, PGR & SEEDS
AVAILABLE IN THE WORLD FOR ALL CROPS
TO INDIAN FARMERS WITH INTERNATIONAL STANDARDS



STOLE THE MINDS OF
INDIAN FARMERS WITH OUR
WATER MELON & MUSK MELON
SEEDS



pvrkumar@ghanacrops.com

www.ghanacrops.com

Dhana Crop Sciences Limited

Unit 604 A, 6th Floor Jain Sadguru Images Capital Park
Image Gardens Road, Madhapur, Hyderabad-500081

SRIBIO
A Microbiome Company

Smart Microbes for Sustainable Agricultural Productivity

Health to the Soil...Strength to the Crop...Wealth to the Farmer...



Prof. Birbahadur - CENTRE FOR CROP MICROBIOME AND NANO RESEARCH

Sri BioAesthetics Pvt. Ltd.

Plot No. C-49, TSIC, IDA Sulthanpur, General Industrial Park
Ameenpur Mandal, Sangareddy District, Pin - 502314, Telangana State, INDIA.
CustomerCare Manager : 63099 81055,
E-mail: sribioaesthetics@gmail.com, www.sribioaesthetics.com

125
Years of
Bayer in India



Health for All

Hunger for None

 Science for a **better life**



FSII is an association of research based plant science industry committed to deliver quality seeds for farmers.

Alliance for Agri Innovation, a special interest group of FSII, promotes technology innovations in agriculture.

 **FSII_India**

 **india.fsii**

 **www.fsii.in**

 **AllianceAgri**

 **AgriInnovation.india**

 **www.agriinnovation.in**



SVSAV TECHNOLOGIES
innovative thinking

www.svsavtechnologies.com



